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Measurement of $\mathcal{B}(B_c^+ \to J/\psi\,\pi^+)/\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)$

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Abstract

While $B_c^+ \to J/\psi\,\pi^+$ and $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ decays were observed before, their relative rates have not been yet measured. Using a data sample corresponding to an integrated luminosity of $1\,\text{fb}^{-1}$ of pp collisions at centre of mass energy $\sqrt{s} = 7\,\text{TeV}$ collected by the LHCb experiment, we determine $\mathcal{B}(B_c^+ \to J/\psi\,\pi^+)/\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu) = 0.047 \pm 0.003(\text{stat}) \pm 0.005(\text{syst})$, which is consistent only with the lowest predictions among theoretical calculations.

Contents

1 Introduction

The $B_c^\pm$ meson was first discovered at the Tevatron by CDF [1] via the decay channel $B_c^+ \to J/\psi\,\mu^+\nu_\mu$. This channel was subsequently used by CDF [2] and D0 [3] for lifetime measurements. The decay channel $B_c^+ \to J/\psi\,\pi^+$ was also seen by both CDF [4] and D0 [5] and used for mass measurements. We present here the first measurement of a ratio of BRs for hadronic and semileptonic B_c decays. There are many theoretical predictions for BRs of these two channels, and their ratios cover a wide range (see Table 24).

It is also interesting with regards to helping to diagnose cause of tension between theoretical predictions and data for the $B_c^\pm$ meson. At the Tevatron, CDF also performed studies on the production of B_c relative to the lighter B mesons. In particular, they made a measurement of the ratio of cross sections times branching ratios for $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ relative to $B^+ \to J/\psi\,K^+$ at $p\bar{p}$ $\sqrt{s} = 1.96\,\mathrm{TeV}$ in the rapidity range $|\eta| < 1.0$ and with a cut of $p_\mathrm{T} > 4\,\mathrm{GeV}$ [6].[1]

$$\frac{\sigma(B_c)\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)}{\sigma(B)\mathcal{B}(B^+ \to J/\psi\,K^+)} = (29.5 \pm 4.0(stat.)^{+10.7}_{-7.6}(syst.) \pm 3.6(p_\mathrm{T}spec))\% \tag{1}$$

There is a discrepancy when comparing this result to theoretical predictions [7]. In particular, using the PDG [8] branching ratio for $\mathcal{B}(B^+ \to J/\psi\,K^+) = (1.016 \pm 0.033)\times 10^{-3}$, and the theoretical estimates for $\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu) \approx 2 \times 10^{-2}$, one can use Eq. 1 to get $\sigma(B_c)/\sigma(B) \sim 1.5 \times 10^{-2}$. However the theoretical estimates for the ratio of production cross sections are expected to be smaller, on the order of 10^{-3}. The latter prediction is expected to be valid only at very high p_T values, $>$ 30 GeV; power corrections can significantly change this cross section for lower transverse momenta (A. Likhoded, V. Kiselev, A. Luchinsky private communication).

At LHCb a measurement was made of the ratio of cross sections times branching ratios for $B_c^+ \to J/\psi\,\pi^+$ relative to $B^+ \to J/\psi\,K^+$ at pp $\sqrt{s} = 7\,\mathrm{TeV}$ in the rapidity range $2.5 < \eta < 4.5$ and with a cut of $p_\mathrm{T} > 4\,\mathrm{GeV}$ [9].

$$\frac{\sigma(B_c)\mathcal{B}(B_c^+ \to J/\psi\,\pi^+)}{\sigma(B)\mathcal{B}(B^+ \to J/\psi\,K^+)} = (0.68 \pm 0.10(stat.) \pm 0.03(syst.) \pm 0.05(lifetime))\% \tag{2}$$

So far this measurement has not been compared to theory. Since absolute normalization of theoretical prediction is often very uncertain, it would be useful to compare the ratio of the LHCb and CDF B_c cross-section measurements to theoretical prediction of the scaling factor between the Tevatron central region and the forward region at LHC. To make such comparison possible, the ratio of branching fractions between the $B_c^+ \to J/\psi\,\pi^+$ and $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channels,

$$R = \frac{\mathcal{B}(B_c^+ \to J/\psi\,\pi^+)}{\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)} \tag{3}$$

[1] CDF also measured similar ratio for $p_\mathrm{T} > 6$ GeV using both muons and electrons but using much smaller data sample, thus with larger errors [1].

must be known. There is considerable variation in the theoretical predictions of R, see Table 24. In this analysis we determine R experimentally.

2 Data and Monte Carlo Samples

This analysis uses the full $1\,\mathrm{fb}^{-1}$ of 2011 Stripping 17b data, as the analysis was started before full 2012 data set was available. The uncertainties are limited by systematics, so there is no motivation to redo the analysis on a larger data sample. Further, this would require regeneration of many MC samples.

In order to best match the data the MC samples use Reco12a, TCK 0x40760037, and Stripping 17b. The MC consisting of B_c decays were generated using BcVegPy. For the $B_c^+ \to J/\psi\,\pi^+$ channel 2M signal MC events were used, along with 1M $B_c^+ \to J/\psi\,K^+$ events. For the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel we use three different signal MC models generated privately with 1M events each. We also make use of privately generated MC for various feed down channels. For $B_c^+ \to (\psi(2S) \to J/\psi\,X)\mu^+\nu$ feed down: 500k, for $B_c^+ \to (\chi_c \to J/\psi\gamma)\mu^+\nu$ feed down: 300k, for $B_c^+ \to J/\psi(\tau^+ \to \mu^+\nu\nu)\nu$: 450k. For this channel we also use 20M public plus 10M private inclusive $B_{u,d,s} \to J/\psi\,X$ MC events generated with Pythia6 in order to model the background.

One of the signal models used was discovered to have a bug in the EvtGen implementation [10]. We would like to explicitly state that the MC samples we use were regenerated using the corrected form factors. This also applies to the feed down MC where appropriate.

3 Data Selection

We take J/ψ candidates coming from the **Jpsi2MuMuDetached** stripping line. The data selection strategy for $B_c^+ \to J/\psi\,\pi^+$ follows the method previously used in the published analysis of $\mathcal{B}(B_c^+ \to J/\psi\,\pi^+\pi^-\pi^+)/\mathcal{B}(B_c^+ \to J/\psi\,\pi^+)$ [11, 12]. We select $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ candidates with similar cuts and method, though some differences are unavoidable because of the presence of neutrino in the final state. All charged tracks in the selected final states must pass Kullback-Liebler track-clone distance > 5000.

The cuts used to select J/ψ candidates are listed in Table 1.

We then combine J/ψ candidate, constrained to the nominal J/ψ mass, with a charged pion or a muon candidate (t^+) to form a B_c^+ candidate (for $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ the B_c is only partially reconstructed because of the missing neutrino). The cuts applied at this stage are listed in Table 2. Among them, the $\cos(t^+, \mu^\pm) < 0.9999$ cut was done to further suppress clones.

We require that the B_c^+ candidate satisfy the trigger requirements listed in Table 3. We require TOS on the B_c candidates in order to maximize efficiency and limit room for mismodeling in the MC.

There is one further cut which was found to be useful for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel. Direction from the primary vertex to the B_c vertex determines B_c flight direction. The momenta of the J/ψ and μ are then decomposed into components parallel and perpendicular

Table 1: Cuts applied to J/ψ candidates.

Quantity	Stripping	Offline cuts
PIDmu(μ)	> 0	—
$p_{\mathrm{T}}(\mu)$	$> 550\,\mathrm{MeV}$	$> 900\,\mathrm{MeV}$
$\chi^2_{track}(\mu)/NDOF$	< 5	< 4
$M(\mu^+\mu^-)$	$2996.916 - 3196.916\,\mathrm{MeV}$	$3040 - 3140\,\mathrm{MeV}$
$\chi^2_{vtx}(J/\psi)/NDOF$	< 20	< 9.0
$DLS(J/\psi)$	> 3	> 5
$p_{\mathrm{T}}(J/\psi)$	—	$> 1500.0\,\mathrm{MeV}$
$\chi^2_{\mathrm{IP}}(\mu^+)$ OR $\chi^2_{\mathrm{IP}}(\mu^-)$	—	> 9.0

Table 2: Cuts on a bachelor charged track (t^+) combined with the J/ψ candidate The $\cos(t^+, J/\psi)$ cut refers to the cosine of the opening angle between μ and J/ψ in the transverse plane to the beam. The cut on $\tau(B_c)$ for the semileptonic channel refers to the pseudo decay time.

<table>
<tr><th>Quantity</th><th colspan="2">Cut</th></tr>
<tr><td>$B_c \to$</td><td>$J/\psi\,\pi^+$</td><td>$J/\psi\,\mu^+(\nu)$</td></tr>
<tr><td>$p_{\mathrm{T}}(t^+)$</td><td>$> 1000\,\mathrm{MeV}$</td><td>$> 500\,\mathrm{MeV}$</td></tr>
<tr><td>$\chi^2_{track}/NDOF(t^+)$</td><td colspan="2">< 4.0</td></tr>
<tr><td>$\chi^2_{IP}(t^+)$</td><td colspan="2">> 9.0</td></tr>
<tr><td>$\cos(t^+, \mu^\pm)$</td><td colspan="2">< 0.9999</td></tr>
<tr><td>PIDK(t^+)</td><td>< 5</td><td>—</td></tr>
<tr><td>PIDmu(t^+)</td><td>—</td><td>> 0</td></tr>
<tr><td>$\cos(t^+, J/\psi)$</td><td colspan="2">> -0.8</td></tr>
<tr><td>$\chi^2_{vtx}/NDOF(B_c)$</td><td colspan="2">< 9.0</td></tr>
<tr><td>$\tau(B_c)$</td><td colspan="2">$> 0.25\,\mathrm{ps}$</td></tr>
</table>

to the B_c flight direction. The perpendicular components are added together to form the quantity $\vec{p}^{\perp}_{J/\psi\,\mu} = \vec{p}^{\perp}_{J/\psi} + \vec{p}^{\perp}_{\mu}$. From momentum conservation $p^{\perp}_{J/\psi\,\mu} = -p^{\perp}_{\nu}$. For events with high values of $M(J/\psi\,\mu)$ the neutrino momentum is small. Fig. 1 shows this distribution with the signal MC and background MC fit to the data in the $5300 < M(J/\psi\,\mu) < 6100\,\mathrm{MeV}$ region, where its discriminating power can be seen. A loose cut of $p^{\perp}_{J/\psi\,\mu} < 4000\,\mathrm{MeV}$ is used.

In order to ensure maximum signal efficiency, a signal-to-background likelihood-ratio approach was used for further analysis. For a given variable x_i, the signal and background probability density functions $\mathcal{P}_{sig}(x_i)$ and $\mathcal{P}_{bkg}(x_i)$ were constructed. The variables which were used are:

Table 3: Trigger requirements

BcL0(Muon|DiMuon)Decision$_{\mathrm{TOS}}$
BcHlt1Track(AllL0|Muon)Decision$_{\mathrm{TOS}}$
BcHlt1DiMuonHighMassDecision$_{\mathrm{TOS}}$
BcHlt2DiMuonDetached(Heavy|JPsi)Decision$_{\mathrm{TOS}}$
BcHlt2TopoMu(2|3)BodyBBDTDecision$_{\mathrm{TOS}}$

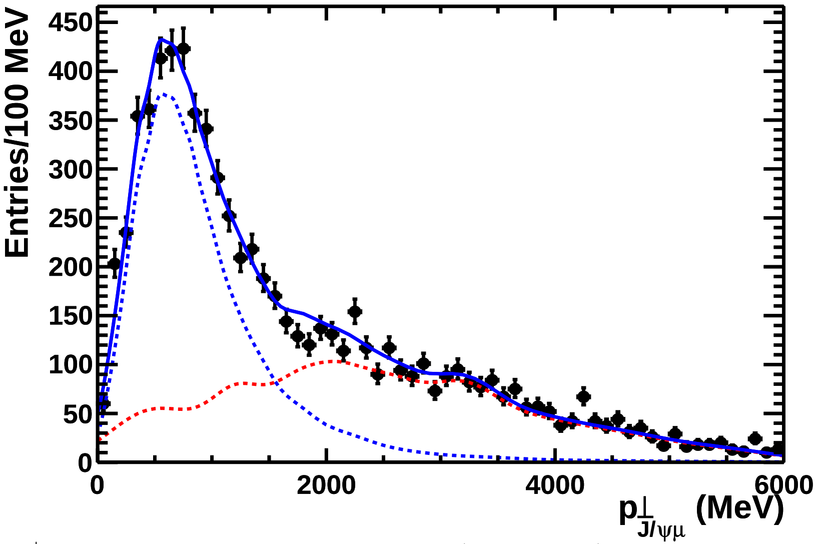


Figure 1: The $p^{\perp}_{J/\psi\,\mu}$ distribution of the signal $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ MC (dashed blue line) plus background MC distribution (dashed red line) fit to the data (black points). The shapes of signal and background distributions were fixed from MC simulations, while their amplitudes were free parameters in the fit.

1. $\chi^2_{IP}(B_c)$: The B_c candidate should point towards Primary Vertex. For $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ pointing is only approximate, but still proves to have considerable discrimination power.

2. $\chi^2_{vtx}(B_c)/NDOF$: Quality of vertex fit used to ensure all selected tracks come from the same particle.

3. $\cos(t^+, J/\psi)$: Due to high $p_T(B_c)$ decay products tend to fly in the same transverse direction. Aside from its use in calculating the DLL, a pre-selection cut of $\cos(t^+, J/\psi) > -0.8$ was made.

Another variable is desired for further discriminating power. Due to differences between the channels, the fourth variable is chosen differently for the two channels. For $B_c^+ \to J/\psi\,\pi^+$ we use:

4. $\chi^2_{IP}(\pi)$: Ensures pion isolation from the PV, where a large amount of π's are produced.

The analogous variable $\chi^2_{IP}(\mu)$ for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel is not as useful, as there are less muons produced at the PV's. Thus for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel we use:

4. $\tau(J/\psi\mu)$ (pseudo decay time): Good for discriminating against short-lived combinatoric background as well as longer-lived background sources.

Using the PDFs built from these variables, the log-likelihood difference between the signal and background hypotheses is calculated:

$$DLL_{sig/bkg} = -2\sum_{i}^{4} \log(\mathcal{P}_{sig}(x_i)/\mathcal{P}_{bkg}(x_i)) \tag{4}$$

To build the signal PDFs for $B_c^+ \to J/\psi\,\pi^+$, a Monte Carlo simulation of $B_c^+ \to J/\psi\,\pi^+$, $J/\psi \to \mu^+\mu^-$ is used. The background PDF's are built from the data with an invariant mass $M(J/\psi\,\pi)$ in the range of $5350 - 5800\,\mathrm{MeV}$ or $6800 - 8500\,\mathrm{MeV}$ (B_c far sidebands). The signal fit region and sidebands are defined as seen in Fig. 2, where the $M(J/\psi\,\pi)$ distribution of the data is shown after preselection cuts.

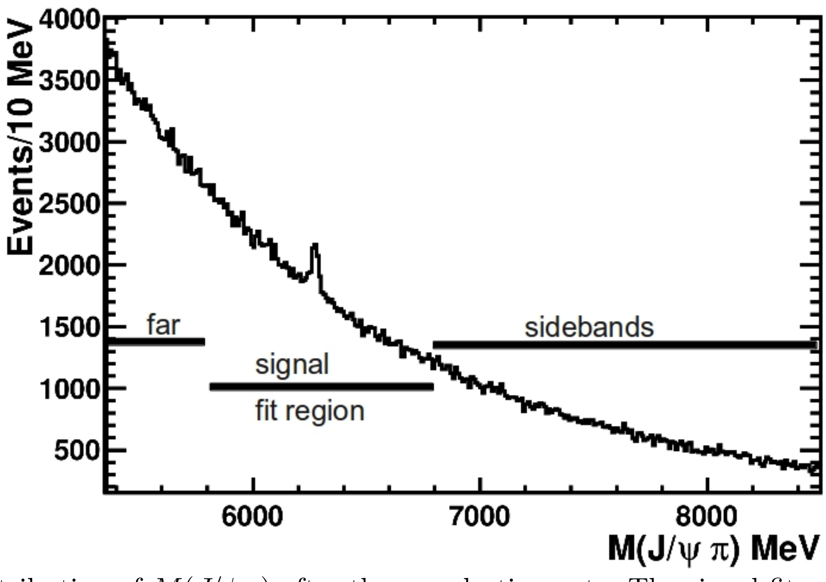


Figure 2: Distribution of $M(J/\psi\,\pi)$ after the preselection cuts. The signal fit region around the B_c mass, and the far sidebands are indicated.

The DLL cut value for the $B_c^+ \to J/\psi\,\pi^+$ channel was found by maximizing the quantity $r = \mathrm{S}/\sqrt{\mathrm{S}+\mathrm{B}}$, where S is the signal yield and B is the background yield within $\pm 2\sigma$ of the B_c peak in $M(J/\psi\,\pi)$. To avoid bias on the signal statistics, S and B dependence on DLL was taken from the signal simulations and far-sidebands, respectively. Absolute normalisation of S(DLL) and B(DLL) was taken from the fit to $M(J/\psi\,\pi)$ distribution in the data in the signal region, for DLL< 0. A plot of r for different DLL cuts is shown in Fig. 3. A cut of $DLL < 1$ was chosen.

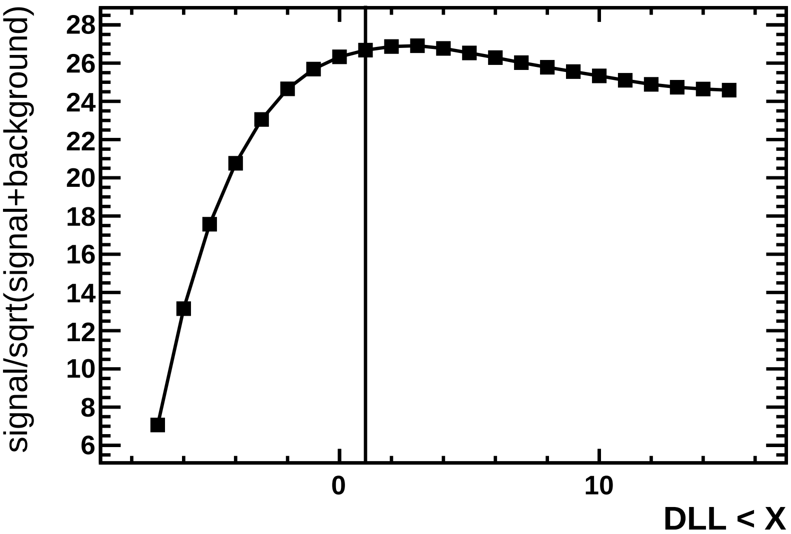


Figure 3: $B_c^+ \to J/\psi\,\pi^+$ Channel: Expected dependence of $\mathrm{S}/\sqrt{\mathrm{S+B}}$ on a DLL cut of $DLL < X$, where X is given on the horizontal axis. The vertical line indicates the actual requirement used in the analysis.

For the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel, the region used to construct the PDFs for both $\mathcal{P}_{sig}(x_i)$ and $\mathcal{P}_{bkg}(x_i)$ was chosen to be in the invariant mass $M(J/\psi\,\mu)$ range of $5300 - 6100\,\mathrm{MeV}$. The $\mathcal{P}_{sig}(x_i)$ PDFs were built using events from signal MC, and the $\mathcal{P}_{bkg}(x_i)$ PDFs were built using events from inclusive background MC consisting of a mixture of $B_{u,d,s} \to J/\psi\,X$ decays. It is a deliberate strategic choice throughout the analysis to remain in the region $M(J/\psi\,\mu) > 5300\,\mathrm{MeV}$. By doing so, we stay above the kinematically allowed region of other B decays. Thus background events in the signal region are largely going to consist of a real μ and a real J/ψ coming from different decays. At lower masses, the backgrounds will become dominated by decays of the form $B \to J/\psi\,$+hadron(s), with a hadron faking the μ. This can be seen in Fig. 4, where the signal and background $M(J/\psi\,\mu)$ distributions are shown with the normalizations determined from the fit to the data. Just below the signal region the large $B \to J/\psi\,K$ peak is seen, and below that contributions from higher body decays. These backgrounds are much more difficult to model (muon fake rates; not all branching ratios for $B \to J/\psi\,$+hadrons are known), and it is highly desirable to avoid them. Furthermore, the $M(J/\psi\,\mu) > 5300\,\mathrm{MeV}$ requirement makes feeddown from semileptonic decays of B_c to heavier charmonium states small. The reason can be seen in Fig. 15, where it is seen that the missing energy from unreconstructed particles push the mass distribution towards lower values.

For the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ signal MC, three different models were used. The models are henceforth referred to as the Kiselev [13–15], Ebert [16], and ISGW2 [17] models. The Kiselev paper calculates the form factors in the framework of the QCD sum rules approach at $q^2 = 0$ and use a pole dependence to extend through the full kinematic range. In the

Ebert et al paper, the q^2 dependence of the form factors in the whole kinematical range are calculated in the framework of the relativistic quark model based on the quasipotential approach. The ISGW2 model is meanwhile based off a nonrelativistic quark model. In Fig. 5 the $M(J/\psi\,\mu)$ distributions are shown for all three signal MC models, where it is seen there is not much variation between them.

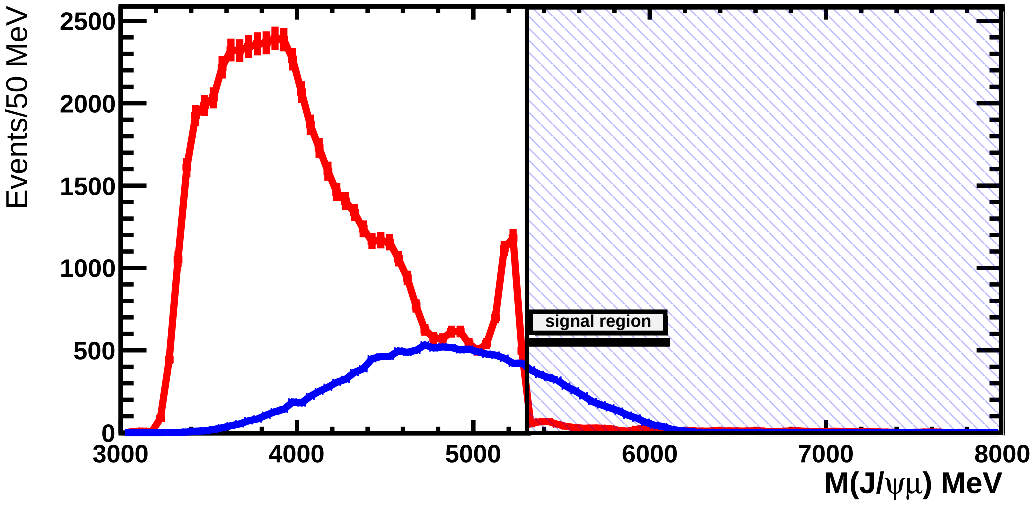


Figure 4: $M(J/\psi\,\mu)$ distributions of the signal MC (blue) and background MC (red) with normalizations determined from the fit to the data. The hatched blue area shows the $M(J/\psi\,\mu)$ region which is to be fit.

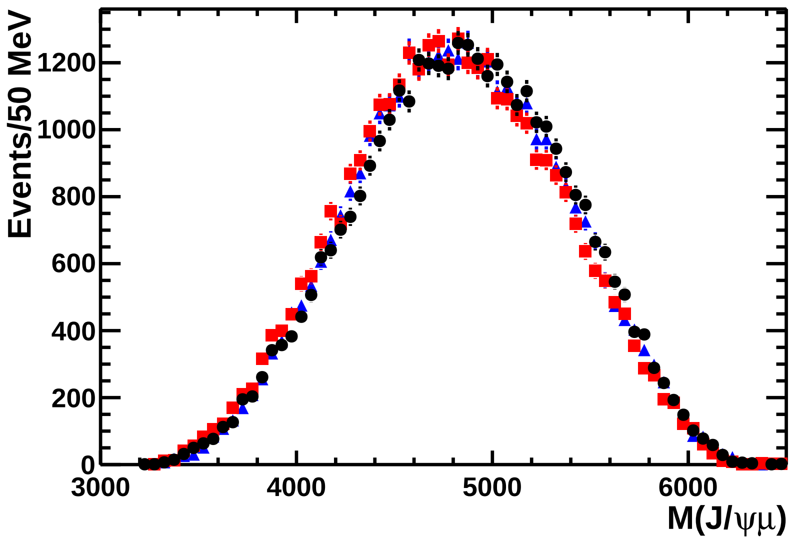


Figure 5: $M(J/\psi\,\mu)$ distribution for the ISGW2 (black circles), Ebert (red squares), Kiselev (blue triangles) signal models.

The DLL distributions of the three theoretical $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ models are shown in

Fig. 6 along with the DLL distribution of the background MC, which demonstrates the ability to discriminate between signal and background events. Furthermore, there is no significant model dependence seen. As it is generally a good midpoint between the studied models, we pick the Kiselev model as nominal model and use the other two as part of the systematic error assessment. For final background suppression, a DLL cut was again chosen by maximizing the quantity $r = \mathrm{S}/\sqrt{\mathrm{S+B}}$. The S and B dependence were taken from the signal and background MC, respectively, in the mass range $5300 < M(J/\psi\,\mu) < 6100\,\mathrm{MeV}$. The absolute normalizations of $S(DLL)$ and $B(DLL)$ were taken from the fit to the $M(J/\psi\,\mu)$ distribution in the data after a cut of $DLL < 0$. A plot of r for different DLL cuts is shown in Fig. 7, and a cut of $DLL < 0$ is thus required in the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel.

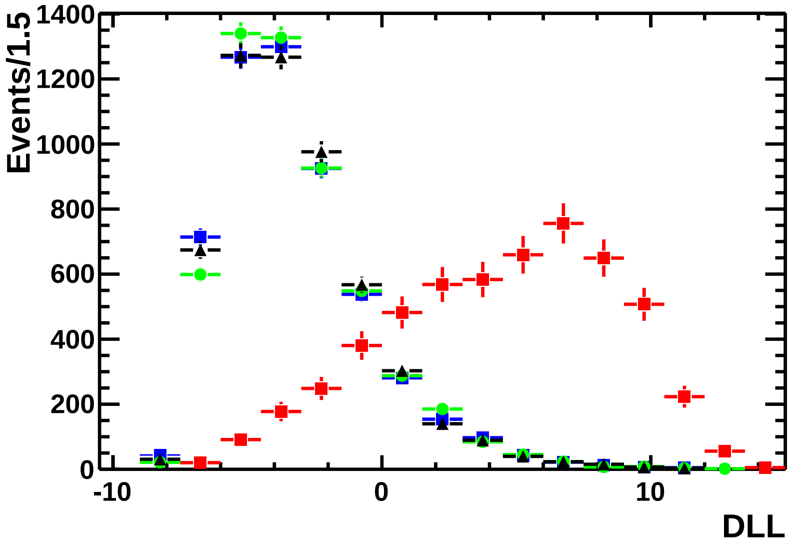


Figure 6: DLL distribution for the three theoretical $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ signal models used (ISGW2- green dots with circles, Ebert- black dots with triangles, Kisilev- blue dots with squares, and background MC- red dots with squares).

We estimate the number of multiple candidate events selected by events which have identical runNumber and eventNumber. We thus make an estimate that for the $B_c^+ \to J/\psi\,\pi^+$ channel 0.12% of events in the fit region are multiple candidate events, while for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel 1.87% of events are multiple candidate events. We conclude that multiple candidate events are small and will be handled in the systematic errors.

4 Signal Monte Carlo Validation

4.1 DLL

To check that the signal MC simulation accurately predicts distributions of the variables used in the log-likelihood formalism, the efficiency for a range of cuts on the individual

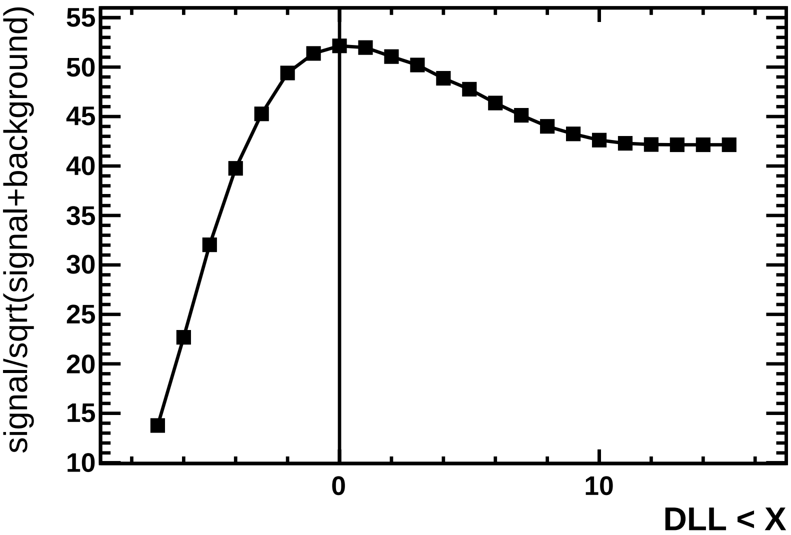


Figure 7: $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ Channel: Expected dependence of $\mathrm{S}/\sqrt{\mathrm{S+B}}$ on a DLL cut of $DLL < X$, where X is given on the horizontal axis. The vertical line indicates the actual requirement used in the analysis.

variables from the MC and data were compared. For example, the behavior of the data and MC under cuts such as $\chi^2_{IP}(B_c) < X$ for different X was studied. The efficiency here is defined as a relative efficiency to the efficiency after the preselection cuts (i.e. efficiency after preselection is 100% by definition). It was observed that the resolution of the MC was too optimistic for the $\chi^2_{vtx}/NDOF(B_c)$ and $\chi^2_{IP}(B_c)$ variables. To fix this, the MC distributions for these variables were smeared to bring the efficiencies into better agreement. For details on the procedure and to see the full set of efficiency comparisons, see in the Appendix Sec 12.1.

The efficiency comparison for the full DLL variable in the $B_c^+ \to J/\psi\,\pi^+$ channel is shown in Fig. 8. The efficiency comparison for the DLL variable in the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ can be seen in Fig. 9. Overall there is quite good agreement between the MC and data for the individual DLL-input variables used as well as for the full DLL variable.

One more variable which required checking is $\tau(B_c)$. As a reminder this variable is only used in the DLL calculation for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel. However due to the greater relative simplicity of studying this variable in the $B_c^+ \to J/\psi\,\pi^+$ channel, it is first studied there. Shown in Fig 10 is the $\tau(B_c)$ distribution for background-subtracted data along with the signal MC distribution, which at the time of DLL construction used the PDG [8] value for the B_c lifetime. The average visible lifetime for this signal MC is 17% smaller than the average visible lifetime for the background-subtracted data. Use of Monte Carlo in the construction of DLL which disagrees with the data leads to less than optimal data selection. Therefore, when constructing DLL we scaled effective (i.e. pseudo-proper) $\tau(B_c)$ for $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ events in MC up by the factor determined in in the $B_c^+ \to J/\psi\,\pi^+$ channel. Recently, dedicated LHCb lifetime measurement using $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ events found $\tau(B_c) = 509 \pm 8(\mathrm{stat.}) \pm, 12(\mathrm{syst.})$ fs [18] which is (13 ± 3)% larger than the PDG [8]

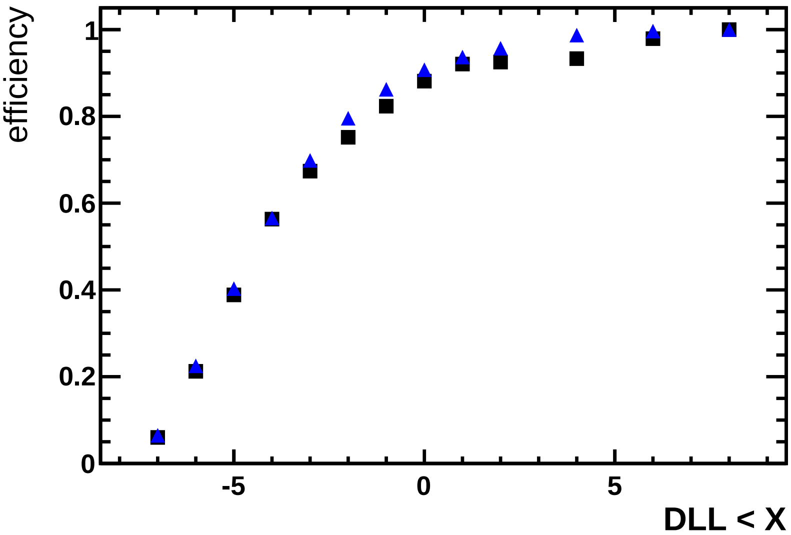


Figure 8: Fraction of $B_c^+ \to J/\psi\,\pi^+$ candidates after the preselection passing a cut of $DLL < X$, where X is given on the horizontal axis, for signal Monte Carlo (blue triangles) and signal in the data (black squares).

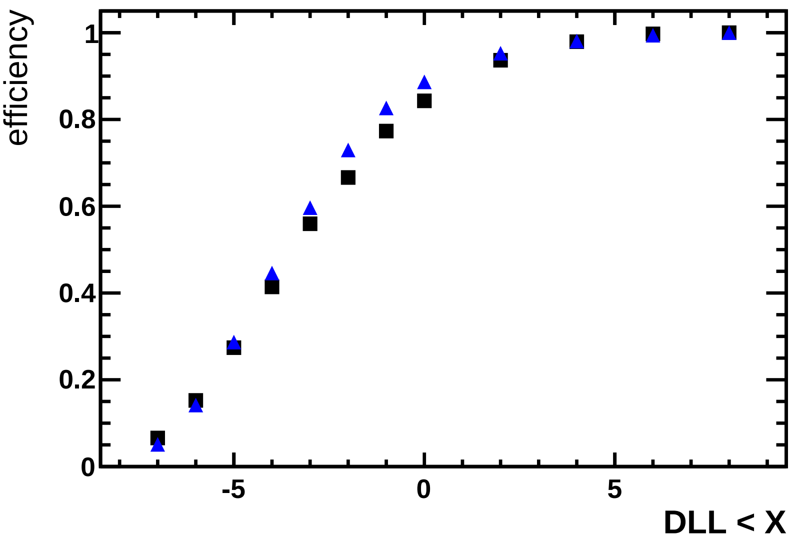


Figure 9: Fraction of $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ candidates after the preselection passing a cut of $DLL < X$, where X is given on the horizontal axis, for signal Monte Carlo (blue triangles) and signal in the data (black squares).

lifetime value, 452 ± 33 fs, assumed in the default MC. This explains and validates our rescaling factor.

Including the new LHCb measurement, the new world average becomes 500 ± 13 fs. The MC used for efficiency calculations is reweighted according to $\exp(-\tau_d/\tau_n)/\exp(-\tau_d/\tau_{MC})$ where τ_d is the proper decay time of the particular event extracted from the truth information, τ_n is the new lifetime to which the MC is being reweighted, and τ_{MC} is the lifetime the MC was generated with.

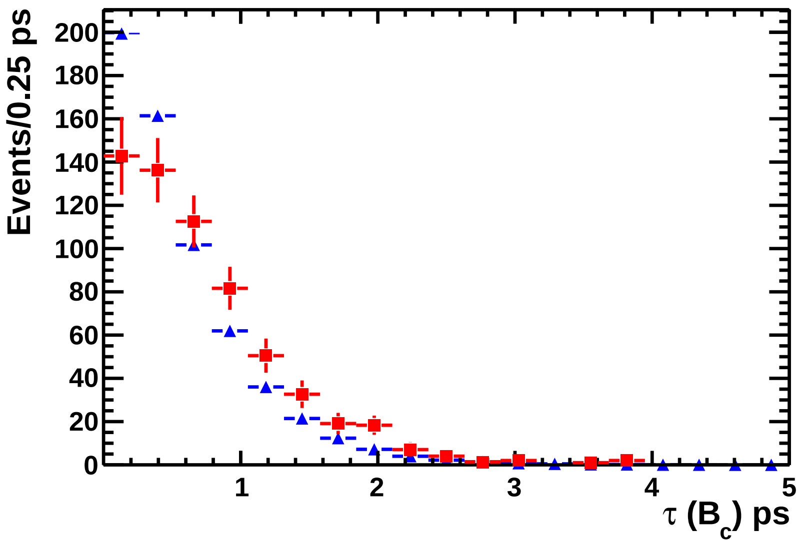


Figure 10: Comparison of $\tau(B_c)$ distribution in the $B_c^+ \to J/\psi\,\pi^+$ channel for background-subtracted data (red squares) and signal MC (blue triangles)

Our initial attempt to model the DLL background shape in the signal region for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel was to use the upper $M(J/\psi\,\mu)$ sideband (beyond the kinematic limit for B_c decays). This turned out to be a poor model because of $M(J/\psi\,\mu)$ dependence. However, it was seen that the upper sideband of the background MC reproduces the DLL distribution of the data in this region well as shown in Fig. 11 which displays the DLL distributions after all other cuts. This gives us a confidence that the background MC can model DLL background shape, and therefore, its use in the $M(J/\psi\,\mu)$ signal region is justified.

5 Extraction of $B_c^+ \to J/\psi\,\pi^+$ signal yield

The $B_c^+ \to J/\psi\,\pi^+$ signal yield is extracted from fitting the $M(J/\psi\,\pi)$ distribution. A double-sided Crystal Ball function (Eq. 5) was used to describe the signal shape. Other signal shape functions were examined in systematic studies. A fit to the MC is shown in Fig. 12. A summary of the parameters returned by the fit to the MC is shown in Table 4. In the fit to the data, the tail parameters α and n were fixed by the fit to the MC, while M_0 and σ were free.

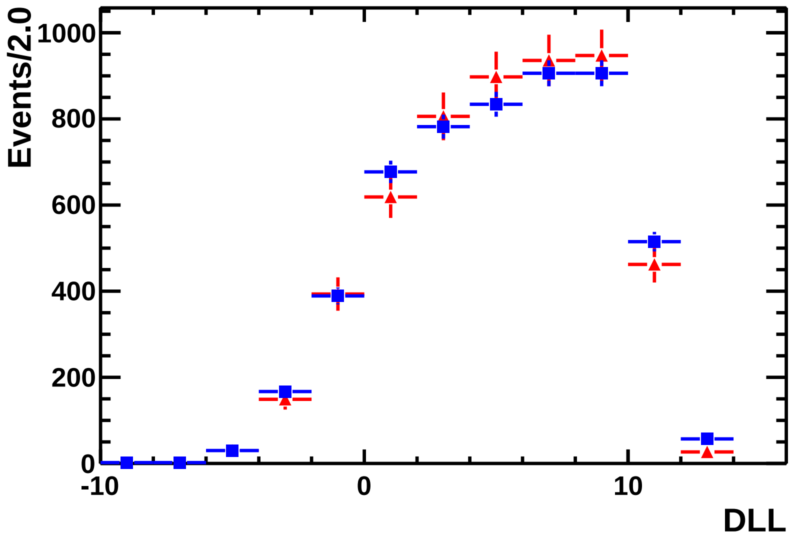


Figure 11: DLL distribution for the background MC (red dots with triangles) and data (blue dots with squares) in $M(J/\psi\,\mu)$ upper sideband ($6300-8500\,\mathrm{MeV}$).

$$\mathrm{DSCB}(M;M_0,\sigma,\alpha_l,n_l,\alpha_r,n_r)=\begin{cases}\dfrac{\left(\frac{n_l}{|\alpha_l|}\right)^{n_l}e^{-\frac{1}{2}\alpha_l^2}}{\left(\frac{n_l}{|\alpha_l|}-|\alpha_l|-\frac{M-M_0}{\sigma}\right)^{n_l}} & \text{if } M-M_0<-|\alpha_l\sigma| \\ \exp\left(-\frac{1}{2}\left(\frac{M-M_0}{\sigma}\right)^2\right) & \text{if } -|\alpha_l\sigma|<M-M_0<|\alpha_r\sigma| \\ \dfrac{\left(\frac{n_r}{|\alpha_r|}\right)^{n_r}e^{-\frac{1}{2}\alpha_r^2}}{\left(\frac{n_r}{|\alpha_r|}-|\alpha_r|+\frac{M-M_0}{\sigma}\right)^{n_r}} & \text{if } M-M_0>|\alpha_r\sigma|\end{cases} \tag{5}$$

Table 4: DSCB parameter results for fit to $M(J/\psi\,\pi)$ in the $B_c^+\to J/\psi\,\pi^+$ signal MC.

M_0	6276.13 ± 0.04
σ	11.58 ± 0.04
α_l	1.69 ± 0.02
α_r	1.65 ± 0.03
n_l	2.73 ± 0.13
n_r	8.58 ± 0.85

The background is modeled with an exponentially decaying shape. However we also need to consider whether it is necessary to account for the peaking contribution from the Cabibbo suppressed mode $B_c^+\to J/\psi\,K^+$, whose first observation was reported by LHCb [19] along with the measurement $\frac{\mathcal{B}(B_c\to J/\psi\,K)}{\mathcal{B}(B_c\to J/\psi\,\pi)}=0.069\pm0.019\pm0.005$. To accomplish this, a MC simulation for this channel was passed through an identical selection as the

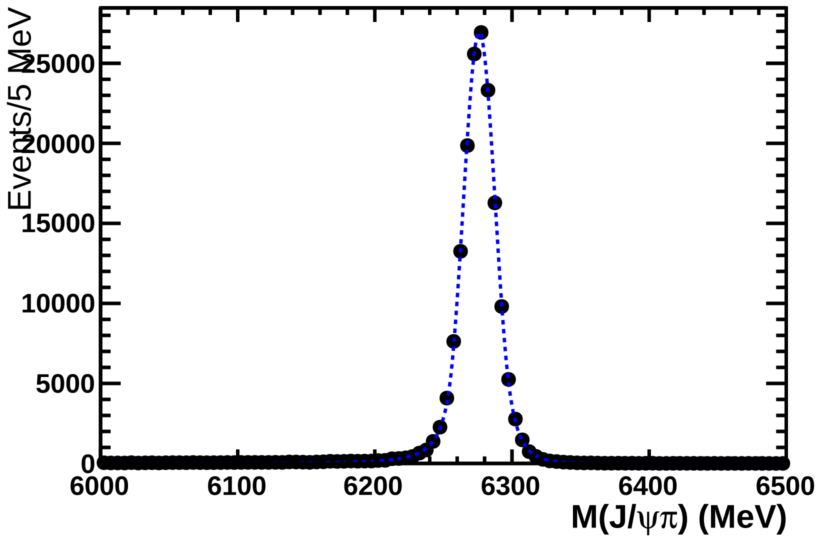


Figure 12: Fit of $M(J/\psi\pi)$ distribution in $B_c^+ \to J/\psi\pi^+$ signal MC with a double-sided Crystal Ball distribution

$B_c^+ \to J/\psi\pi^+$ channel. To describe the $B_c^+ \to J/\psi K^+$ component, a double-sided Crystal Ball function (Eq. 5) is also used. The fit to the MC can be seen in Fig. 13. A summary of the parameters returned by the fit to the MC is shown in Table 5. In the fit to the data the parameters were fixed to these values. From this simulation we also estimate expected ratio of $B_c^+ \to J/\psi K^+$ events to the signal yield via:

$$\frac{N_{J/\psi K}}{N_{J/\psi\pi}} = \left(\frac{\mathcal{B}(B_c \to J/\psi K)}{\mathcal{B}(B_c \to J/\psi\pi)}\right)\left(\frac{\epsilon_{J/\psi K}}{\epsilon_{J/\psi\pi}}\right) \tag{6}$$

We find the ratio of full efficiencies including acceptance, trigger, and selection to be $\epsilon_{J/\psi K}/\epsilon_{J/\psi\pi} = 0.1402 \pm 0.0013$. In this calculation, we use the PIDCalib package to get the efficiency directly from the data for the $PIDK < 5$ cut applied to the bachelor track. Performing the fit without any component for the $B_c^+ \to J/\psi K^+$ channel yields $N_{J/\psi\pi} = 841.27 \pm 40.40$. From this and Eq. 6 we get an estimate of $N_{J/\psi K} = 8.2 \pm 2.4$ events. When letting $B_c^+ \to J/\psi K^+$ contribution float in the fit, we obtain $N_{J/\psi K} = 76.4 \pm 31.9$. There is a large uncertainty, and this result is unrealistically high. The excess over the expected $B_c^+ \to J/\psi K^+$ yield likely comes from the shape of the background. Thus for the default fit we constrain the number of $N_{J/\psi K}$ events to a fraction of the signal yield according to Eq. 6. The overall effect of the $B_c^+ \to J/\psi K^+$ component is quite small: the fit without the component included differs from the fit with it included by +0.3%.

Thus the total fit is the sum of signal contribution, the exponentially decaying background, and the $B_c^+ \to J/\psi K^+$ background which is fixed to the expected fraction of signal events. We used extended likelihood fit to the unbinned data. The total fit to the data can be seen in Fig. 14. The B_c parameters returned by the fit are shown in Table 6. We observe $N_{J/\psi\pi} = 839.0 \pm 40.1$ signal events. As a measure of goodness of fit, we report the fit returns $\chi^2/NDOF = 1.0054$ with a confidence level of 45.69%.

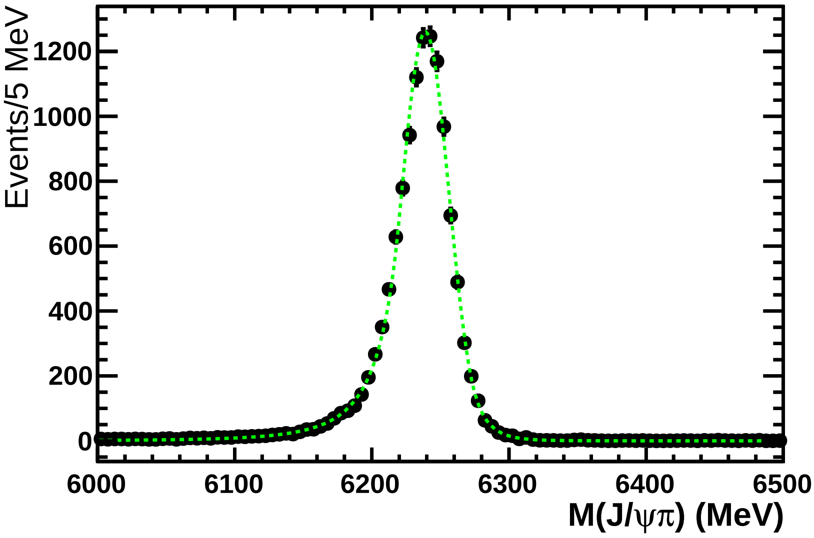


Figure 13: Fit of $M(J/\psi\,\pi)$ distribution in the $B_c^+ \to J/\psi\,K^+$ MC with an asymmetric, double-sided Crystal Ball distribution

Table 5: DSCB parameter results for fit to $M(J/\psi\,\pi)$ in the $B_c^+ \to J/\psi\,K^+$ MC

σ	17.27 ± 0.30
M_0	6239.00 ± 0.28
α_l	1.13 ± 0.06
α_r	1.86 ± 0.11
n_l	4.03 ± 0.57
n_r	9.65 ± 9.14

6 Extraction of $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ signal yield

6.1 Feed Down Considerations

For the decay channel $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ it is necessary to consider feed down from B_c semileptonic decays to other charmonium states. In particular how it can be expected to deform the $M(J/\psi\,\mu)$ distribution. To start, we discuss how much feed down to expect. The feed down fraction from $B_c^+ \to \psi_{feed}\mu\nu$ can be estimated from:

$$\alpha_{feed} = \frac{N_{feed}}{N_{sig}} = \left(\frac{\epsilon_{feed}}{\epsilon_{sig}}\right)\left(\sum_X \mathcal{B}(\psi_{feed} \to J/\psi\,X)\right)\left(\frac{\mathcal{B}(B_c \to \psi_{feed}\mu\nu)}{\mathcal{B}(B_c \to J/\psi\,\mu\nu)}\right)$$

We first consider feed down from $B_c^+ \to \psi(2S)\mu^+\nu_\mu$, with the $\psi(2S)$ decaying to one of the channels with a J/ψ in the final state as summarized in Table 7. We calculate $\alpha_{\psi(2S)}$, which is defined as in Eq 7 for $\psi_{feed} = \psi(2S)$.

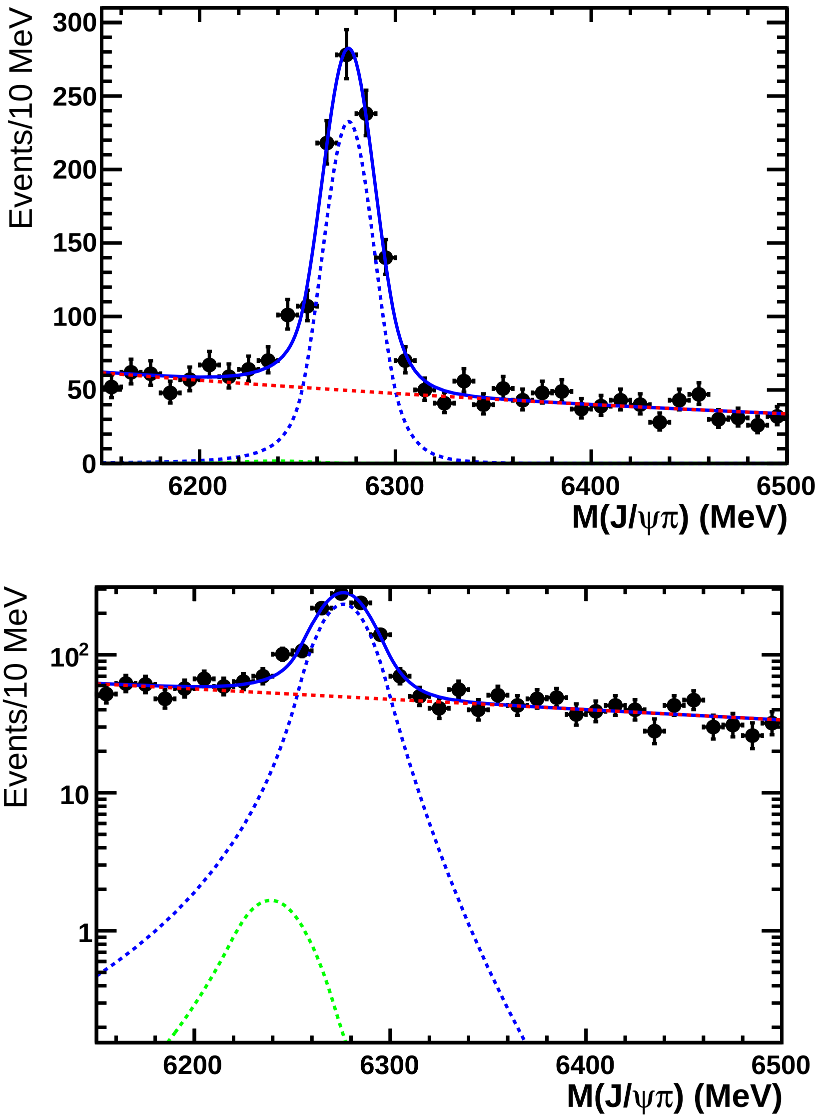


Figure 14: Full fit to the $M(J/\psi\,\pi)$ distribution of the data with signal (dashed blue), $B_c^+ \to J/\psi\,K^+$ (dashed green), and exponentially decaying background (dashed red) contributions shown.

To study selection efficiency for this channel we generated $B_c^+ \to \psi(2S)\mu^+\nu_\mu$ MC, with $\psi(2S)$ decaying according to the branching fractions listed in this table.

The feed down MC sample was then passed through an identical selection as the data. It was found that the feed down MC gives a significantly lower efficiency than the signal MC. The source of this can be seen in Fig. 15, where it is shown that the additional missing particles in the feed down decays push $M(J/\psi\,\mu)$ into relatively low mass ranges. The mass cut alone ($5300 - 8000$ MeV) then makes the feed down efficiency to be quite small. Passed through all the cuts, $\epsilon_{feed}/\epsilon_{sig} = 0.1185 \pm 0.0040$. The theoretical estimate for ratio of branching fractions for $\mathcal{B}(B_c \to \psi(2S)\mu\nu)/\mathcal{B}(B_c \to J/\psi\,\mu\nu)$ range from $0.0092 - 0.185$, and

Table 6: B_c parameters returned by fit to the $M(J/\psi\,\pi)$ distribution of data

σ	13.53 ± 0.73
M_0	6276.13 ± 0.70
$N_{J/\psi\,\pi}$	839.0 ± 40.1

Table 7: Possible feed down channels and their PDG [8] branching ratios.

Channel	BR
$\psi(2S) \to J/\psi\,\pi^+\pi^-$	$(33.6 \pm 0.4)\%$
$\psi(2S) \to J/\psi\,\pi^0\pi^0$	$(17.75 \pm 0.34)\%$
$\psi(2S) \to J/\psi\,\eta$	$(3.28 \pm 0.07)\%$
$\psi(2S) \to J/\psi\,\pi^0$	$(1.3 \pm 0.10) \times 10^{-3}$
$\psi(2S) \to \gamma(\chi_c \to \gamma J/\psi)$	$(5.0 \pm 0.2)\%$
$\sum_X \mathcal{B}(\psi(2S) \to J/\psi\,X)$	$(59.76 \pm 0.56)\%$

are listed in Table 8. Taking the midpoint of the theoretical predictions, $\alpha_{\psi(2S)} = 0.69\%$.

Table 8: Different theoretical estimates of $\frac{\mathcal{B}(B_c^+ \to \psi(2S)\mu^+\nu_\mu)}{\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)}$. For KLL model two values are given corresponding to different models of $\psi(2S)$ wave function.

Kiselev [13]	Ebert [16]	ISGW2 [17]	CC [20]	CF [21]	LC [22]	KLL [23]
0.0495	0.0252	0.185	0.0422	0.0800	0.0092	0.164 (0.0904)

The next contribution we consider is from $B_c \to (\chi_{cJ} \to \gamma J/\psi)\mu\nu$. We again calculate α_{χ_c}, which is defined as in Eq 7 for $\psi_{feed} = \chi_c$. The theoretical predictions for the ratio of branching fraction for this feed down mode, summed over different J, to the signal channel are given in Table 9. We use PDG [8] vales for $\mathcal{B}(\chi_{cJ} \to \gamma J/\psi)$. The midpoint of theoretical predictions gives $\sum_J \mathcal{B}(\chi_{cJ} \to \gamma J/\psi) \times \mathcal{B}(B_c \to \chi_{cJ}\mu\nu)\mathcal{B}/(B_c \to J/\psi\,\mu\nu) = 3.5\%$. The unreconstructed photons can be expected to again lower the efficiency by pushing the $M(J/\psi\,\mu)$ distribution out of the signal region. To estimate the efficiency, we implemented theoretical form-factors [24] for $B_c \to \chi_c\mu\nu$ in the EvtGen code. Again, the events were passed through an identical selection as the data. From this, we estimate $\epsilon_{feed}/\epsilon_{sig} = 0.3636 \pm 0.0093$. Using the theory predictions and measured branching fractions results in $\alpha_{\chi_c} \approx 1.27\%$.

There is also feed down to be considered from $B_c^+ \to J/\psi(\tau^+ \to \mu^+\nu_\mu\overline{\nu}_\tau)\nu_\tau$. We calculate the fraction of feed down events for this channel α_τ in the same manner as

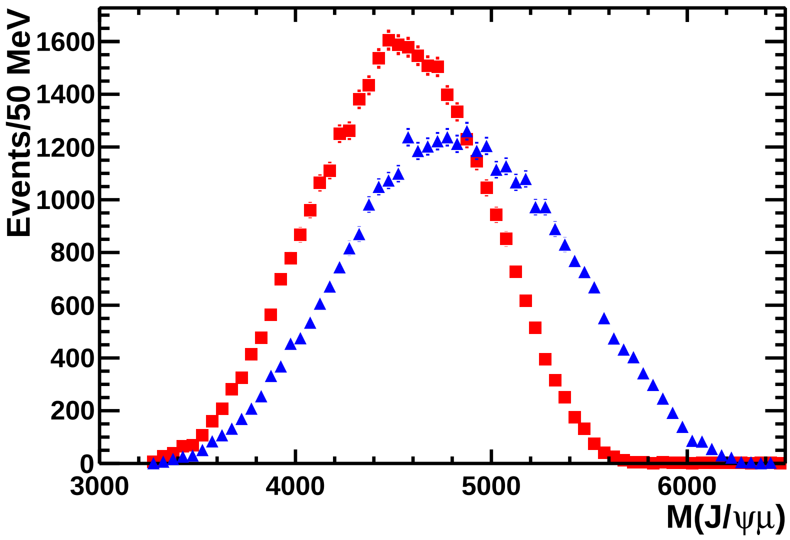


Figure 15: $M(J/\psi\,\mu)$ distribution of signal MC (blue triangles) and feed down (red squares) from $B_c^+ \to \psi(2S)\mu^+\nu_\mu$, with the $\psi(2S)$ decaying to a channel with J/ψ in the final state.

Table 9: Different theoretical estimates of $\sum_J \mathcal{B}(\chi_{cJ} \to \gamma J/\psi) \times \frac{\mathcal{B}(B_c \to \chi_{cJ}\mu\nu)}{\mathcal{B}(B_c \to J/\psi\,\mu\nu)}$.

IKS [25]	CCWZ [26]	HNV [27]
0.032	0.038	0.032

Eq 7. This decay can be expected to contribute little to the number of signal-like events selected. The two neutrinos from the τ^+ will carry away a large amount of energy due to the large τ^+ mass. From a full simulation of these events, we get $\epsilon_{feed}/\epsilon_{sig} = 0.0141 \pm 0.0013$. The branching ratios also need to be folded into the feeddown estimate. PDG [8] gives $\mathcal{B}(\tau^+ \to \mu^+\nu_\mu\overline{\nu}_\tau) = (17.41 \pm 0.04)\%$. Some theoretical predictions for $\mathcal{B}(B_c^+ \to J/\psi\,\tau^+\nu_\tau)/\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)$ are given in Table 10. Together we expect a fraction $\alpha_\tau \approx 0.06\%$ of events to come from $B_c^+ \to J/\psi\,(\tau^+ \to \mu^+\nu_\mu\overline{\nu}_\tau)\nu_\tau$ decays.

Table 10: Different theoretical estimates of $\frac{\mathcal{B}(B_c^+ \to J/\psi\,\tau^+\nu_\tau)}{\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)}$.

IKS [25]	Kiselev [13]	AKNT [28]	HNV [27]
0.237	0.253	0.283	0.266

Another source which was investigated is $B_c^+ \to (B_{d,s} \to J/\psi\,X)\mu^+\nu_\mu$, and similarly for feed down from the $B_{d,s}^*$ states. We implemented form factors from the Kiselev [13] paper in EvtGen for the decay $B_c^+ \to B_s\mu^+\nu_\mu$. The $B_s \to J/\psi\,X$ part of the decay was done according to the inclusive $B_s \to J/\psi X$ MC (Event Type: 13442001). Shown in Fig. 16 is the $M(J/\psi\,\mu)$ distribution at the generator level. We generated 300k events, none of which were in the mass range of $M(J/\psi\,\mu) > 5300$ MeV. Therefore, such feeddown can be safely neglected.

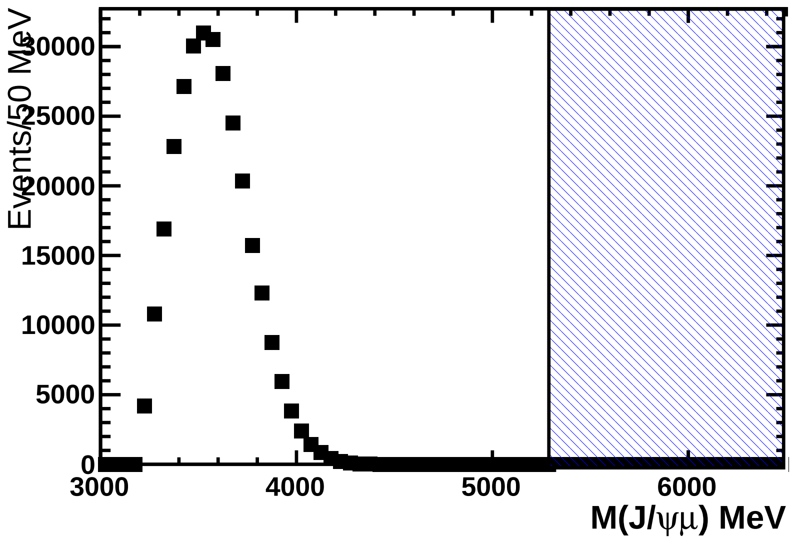


Figure 16: Generator level $M(J/\psi\,\mu)$ distribution of feed down from $B_c^+ \to (B_s \to J/\psi X)\mu^+\nu_\mu$ decays. Shaded in blue is the signal region of $5300 - 6100\,\mathrm{MeV}$.

Total expected ratio of feeddown to signal events is $\alpha = \alpha_{\psi(2S)} + \alpha_{\chi_c} + \alpha_\tau = 2.02\%$. We use this estimate in further analysis and assign a systematic error based on variation of the theoretical predictions used in the calculations above.Since theoretical spread is substantial it is important to keep α small. This is accomplished by working in the $M(J/\psi\,\mu)$ endpoint region. Growth of feeddown fractions upon lowering the $M(J/\psi\,\mu)$ cut off is shown in Fig. 17.

6.2 b Baryon Backgrounds

While the background is modeled using events from inclusive $B_{u,d,s} \to J/\psi X$ MC, we have also considered backgrounds from b baryon decays. We have studied two Λ_b^0 MC samples to investigate expected background shapes, $\Lambda_b^0 \to (\Lambda \to p\pi)J/\psi X$ [2] (Fig. 18) and $\Lambda_b^0 \to J/\psi\, pK$ (Fig. 19).

In both cases, any peaking backgrounds are below the selected mass range of $M(J/\psi\,\mu) >$ 5300 MeV. The events which lie in the selected region are again combinatorial in origin (i.e. the bachelor muon comes from the semileptonic decay of the other b). Therefore their mass distribution is consistent, within the large statistical errors, with the shape of the combinatorial backgrounds from B mesons. Since many of relevant branching fractions for b baryon decays to J/ψ are not measured, we don't attempt to add b baryons to the B

[2]Event Type: 15442001; X are possible transition products of excited charmonium states to J/ψ.

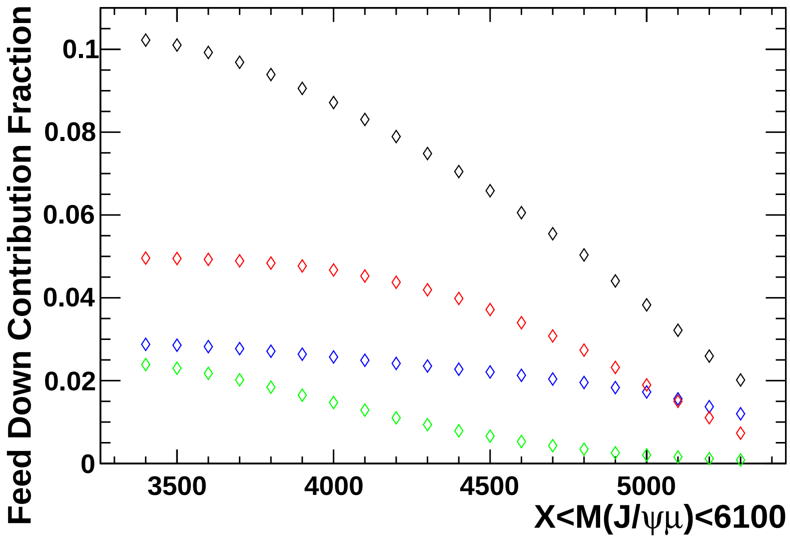


Figure 17: Growth of feed-down fractions as the lower limit of the mass fit window is extended to X: total (black), $B_c \to (\psi(2S) \to J/\psi X)\mu\nu$ (blue), $B_c \to (\chi_c \to \gamma J/\psi)\mu\nu$ (red) and $B_c \to J/\psi(\tau \to \mu\nu\nu)\nu$ (green).

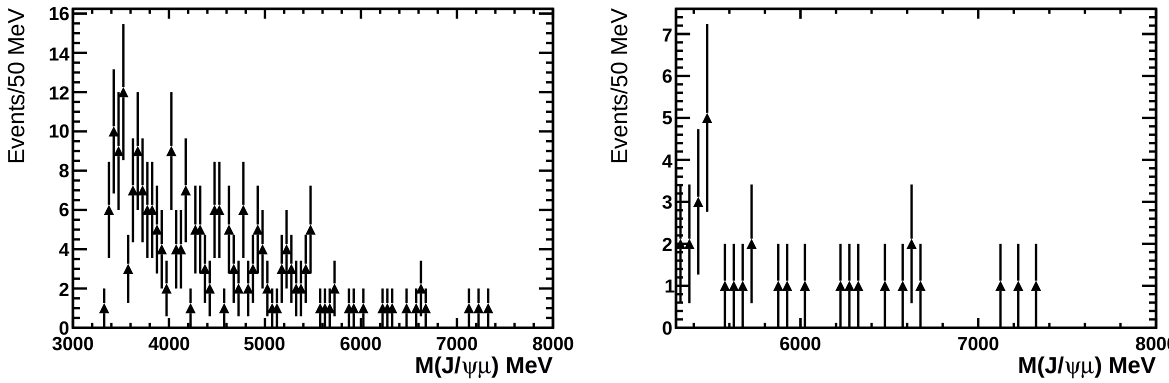


Figure 18: $M(J/\psi\,\mu)$ distributions in the full (left) and selected (right) regions, obtained from simulations of inclusive $\Lambda_b^0 \to (\Lambda \to p\pi)J/\psi\,X$.

mesons in the background simulations. This is justified since the B production rates are larger and the background shapes are consistent.

6.3 $M(J/\psi\,\mu)$ Fit

The signal yield for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel is extracted by fitting the $M(J/\psi\,\mu)$ distribution in the mass range $5300.0 < M(J/\psi\,\mu) < 8000.0$, which extends beyond kinematically allowed signal range to probe the combinatorial background component. As discussed in the previous section, in addition to the signal and combinatorial background we also expect small contribution from B_c feeddown modes. Our overall PDF model is

$$\text{Model}(n_{J/\psi\,\mu}, n_{bkg}) = n_{J/\psi\,\mu} f_{sig} + n_{bkg} f_{bkg} + \alpha n_{J/\psi\,\mu} f_{feeddown} \tag{7}$$

where α is the estimated feeddown fraction discussed in the previous section.

Shapes of each component are parameterized by functions discussed below. While some shape parameters are fixed in the fit to the data, dominant shape parameters are floated in the fit, with the simultaneous constraint to the signal and the background MC samples. In this way, statistical uncertainty in the signal and background MC samples is properly propagated to the extracted signal yield.

We parameterize the signal shape (f_{sig}), as a product of the 3-body phase space distribution ($PS(M(J/\psi\,\mu))$), assuming the third unobserved particle is massless, multiplied by a first order polynomial to allow for dynamical effects:

$$f_{sig}(M(J/\psi\,\mu)|s_1) = PS(M(J/\psi\,\mu)) \quad (1 + s_1\, M(J/\psi\,\mu)) \tag{8}$$

with

$$PS(M(J/\psi\,\mu)) = \left(\frac{m_{B_c}^2 - M(J/\psi\,\mu)^2}{M(J/\psi\,\mu)}\right) \sqrt{(M(J/\psi\,\mu)^2 - (m_{J/\psi} + m_\mu)^2)(M(J/\psi\,\mu)^2 - (m_{J/\psi} - m_\mu)^2)} \tag{9}$$

where the masses $m_{J/\psi}$, and m_μ are set to the PDG [8] values. The parameter m_{B_c} determines an upper cutoff to the $M(J/\psi\,\mu)$ distribution. We do not set it to the known B_c mass, but determine it by the fit to the signal MC to correct for the resolution effects, $m_{B_c} = 6289\,\text{MeV}$. We note that, based off the $M(J/\psi\,\pi)$ fit used in the extraction of the $B_c^+ \to J/\psi\,\pi^+$ signal yield(Sec 5), there is a negligible shift in B_c mass scale between the data and MC. The polynomial coefficient s_1 is floated in the simultaneous fit to the data and to the signal and background MC.

A feeddown shape ($f_{feeddown}$) is parameterized by a similar function:

$$f_{feeddown}(M(J/\psi\,\mu)|d_1, d_2, m_{FD}) = PS(M(J/\psi\,\mu), m_{FD}) \quad (1 + d_1\, M(J/\psi\,\mu) + d_2\, M(J/\psi\,\mu)^2) \tag{10}$$

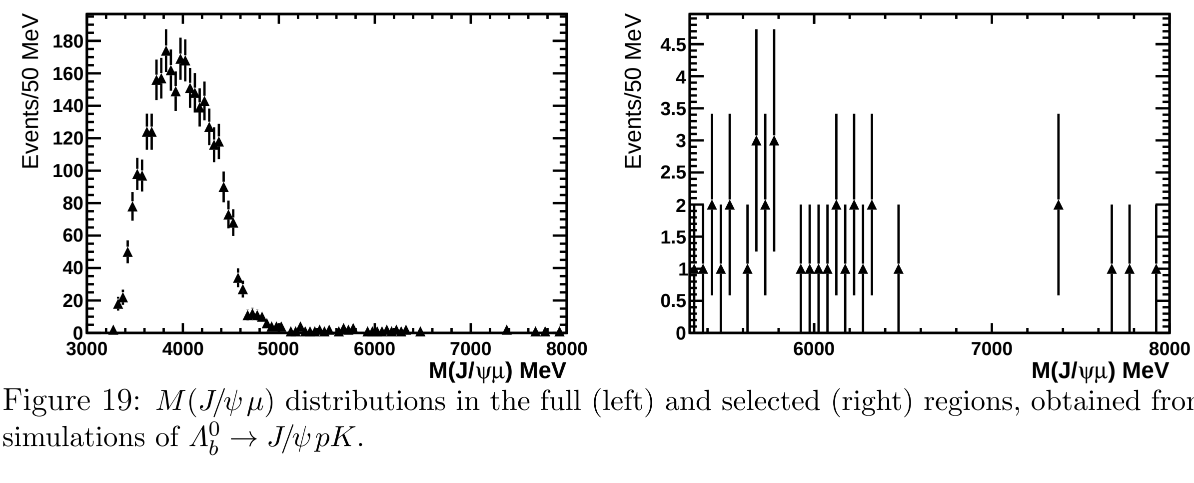


Figure 19: $M(J/\psi\,\mu)$ distributions in the full (left) and selected (right) regions, obtained from simulations of $\Lambda_b^0 \to J/\psi\, pK$.

Here $PS(M(J/\psi\,\mu), m_{FD})$ is the same as Eq 9 but for m_{FD} taking the place of m_{B_c}, i.e. it determines an upper cutoff to the $M(J/\psi\,\mu)$ distribution. It is in contrast left as a free parameter in the fit. Of course m_{FD} doesn't have as clear of a physical significance. This shape is used due to its ability to describe well the distribution and have an analogous form to the signal shape. It is also desirable to have an upper cut off to the mass distribution. The free parameters are fixed by fitting a sum of simulated feeddown channels, normalized by the estimates discussed in the previous section (see Fig. 20). These parameters are then fixed when fitting the data.

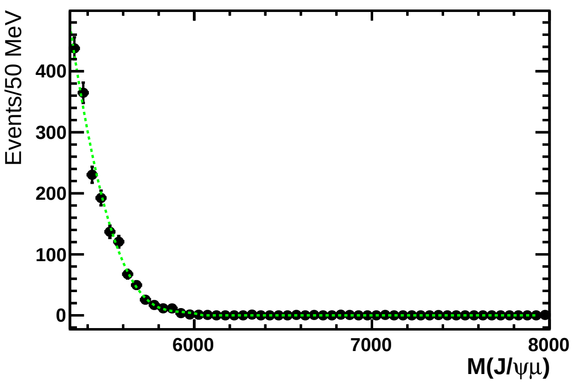


Figure 20: Fit of $f_{feeddown}$ to the feed down MC, with different channels weighted according to estimated contributions discussed in 6.1.

The background shape (f_{bkg}) consists of two components. The dominant form of background is combinatorial, and its shape is taken as $\exp(b_1\bar{M}(J/\psi\,\mu) + b_2\bar{M}(J/\psi\,\mu)^2)$. Where $\bar{M}(J/\psi\,\mu)$ denotes that the mass is shifted as $\bar{M}(J/\psi\,\mu) = M(J/\psi\,\mu) - 5300\,\text{MeV}$. The second component is included in order to account for the upper tail of the $B \to J/\psi\,K$ background peak, and is taken as the tail of a Gaussian. The full background model is parameterized as

$$f_{bkg}(M(J/\psi\,\mu)|b_1, b_2, b_f) = b_f\left(\exp(b_1\bar{M}(J/\psi\,\mu) + b_2\bar{M}(J/\psi\,\mu)^2)\right) + (1-b_f)\text{Gaussian}(M(J/\psi\,\mu)) \tag{11}$$

where b_f is defined as the fraction of the full background model which corresponds to the combinatorial background shape, and is left floating in the fit. Due to inability to trust background MC prediction of fake rates, b_f is floated separately in the fit to the MC and fit to the data. In order to reliably obtain the shape of the peak, the mean value and width of the Gaussian are fixed via a fit to the background MC in the extended range of $M(J/\psi\,\mu) > 5100\,\text{MeV}$. We thus obtain $M_0 = 5196.16 \pm 1.54\,\text{MeV}$ and $\sigma = 45.41 \pm 1.52\,\text{MeV}$.

We perform an unbinned extended likelihood fit. The full fit is superimposed on the data in Fig. 21, and on the the signal and background MC in Fig. 22 and Fig. 23, respectively. The parameters of the fit are given in Table 11, where values without errors are parameters which were fixed before the fit by their respective means described above. We observe $n_{J/\psi\,\mu} = 3536.9 \pm 124.5$ signal events. To quantify goodness-of-fit, we bin the data as shown in the fit displays and calculate a χ^2 value between the data points and the fit function. We obtain $\chi^2/DOF = 0.9076$, corresponding to a confidence level of 65.3%.

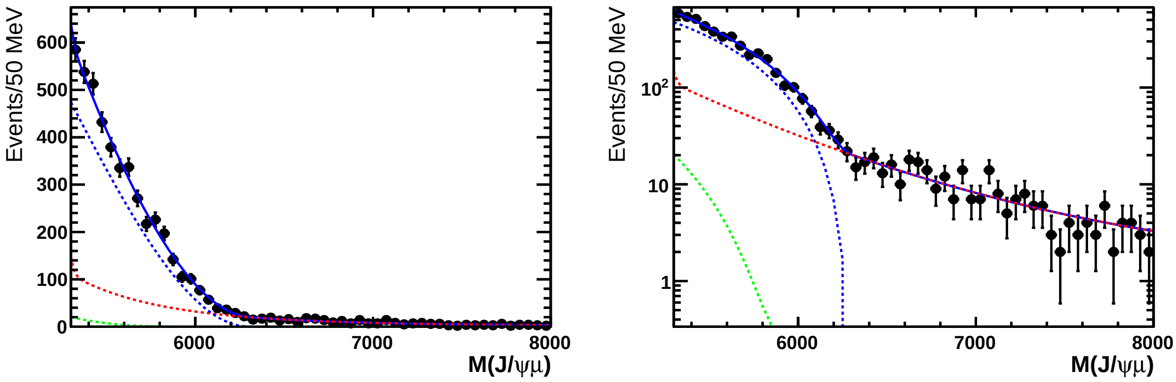


Figure 21: Full fit of the $M(J/\psi\,\mu)$ distribution of the data, with signal (dashed blue), feed down(dashed green), and background (dashed red) components shown. On the right is the same plot with log scale

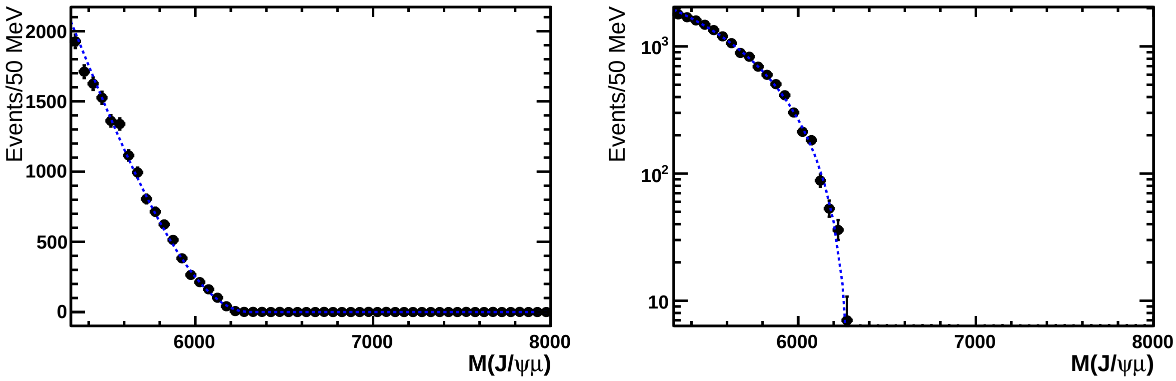


Figure 22: Simultaneous fit result of $f_{sig}(M(J/\psi\,\mu)|s_1)$ to the signal MC with linear and log scale.

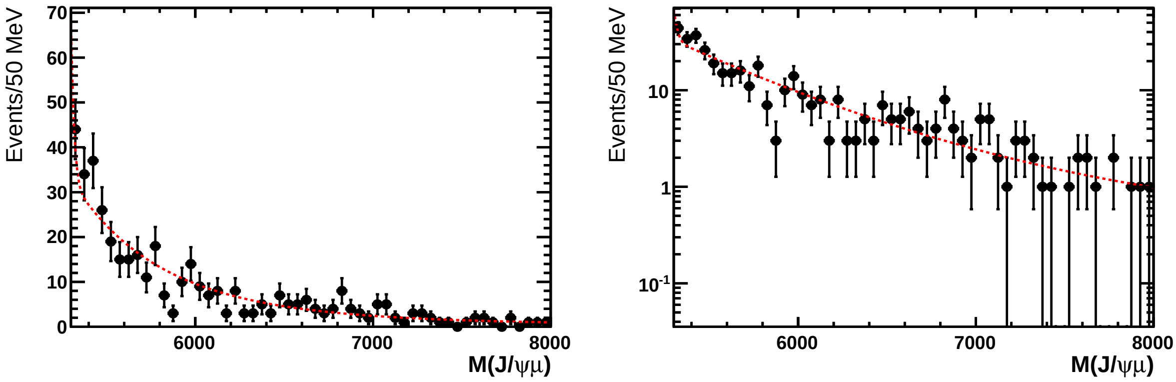


Figure 23: Simultaneous fit result of f_{bkg} to the background MC with linear and log scale.

Table 11: Parameters of simultaneous fit to the $M(J/\psi\,\mu)$ distributions of data and MC. Entries without errors were fixed in this fit by the procedures described in the text.

Signal	$n_{J/\psi\,\mu}$	3536.9 ± 124.5
	s_1	$(-1.584 \pm 0.003) \times 10^{-4}$
	m_{B_c}	6289 MeV
Background	N_{bkg}	1318.5 ± 117.7
	b_1	$(-1.920 \pm 0.216) \times 10^{-3}$
	b_2	$(2.302 \pm 0.784) \times 10^{-7}$
	b_f(Data)	0.989 ± 0.016
	b_f(MC)	0.970 ± 0.016
	σ	45.4 MeV
	M_0	5196 MeV
Feed down	α	0.0202
	m_{FD}	6039.1 MeV
	d_1	-3.402×10^{-4}
	d_2	2.896×10^{-8}

7 Signal Detection Efficiencies

With the exception of the $PIDK < 5$ cut used on the bachelor pion in the $B_c^+ \to J/\psi\,\pi^+$ channel, the MC is used in order to determine the detection efficiencies. The efficiency for the $PIDK$ cut is determined from data using the PIDCalib package.

We factor the total efficiency into $\epsilon_{total} = \epsilon_{acc} \times \epsilon_{sel/reco} \times \epsilon_{trig}$. The efficiency ϵ_{acc} corresponds to the efficiency coming from the geometrical acceptance of the detector. Our MC is generated with the generator level cut BcDaughtersInLHCb, which requires that all charged particles lie within $10\,\text{mrad} < \theta_{\text{charged}} < 400\,\text{mrad}$. There is no cut on the neutrino in the semileptonic channel. The $\epsilon_{reco/sel}$ corresponds to the reconstruction and selection efficiency, while the ϵ_{trig} is the efficiency from the trigger requirements (see Table 3). A summary of the discussed efficiencies and their ratios are listed in Table 12.

Table 12: Table of efficiencies

	acceptance	selection/ reconstruction	trigger	total
$B_c^+ \to J/\psi\,\mu^+\nu_\mu$	0.1390 ± 0.0001	0.0172 ± 0.0001	0.8961 ± 0.0023	$(0.2146 \pm 0.0017) \times 10^{-2}$
$B_c^+ \to J/\psi\,\pi^+$	0.1325 ± 0.0002	0.0991 ± 0.0003	0.8272 ± 0.0008	$(1.0859 \pm 0.0032) \times 10^{-2}$
ratio	1.0491 ± 0.0020	0.1739 ± 0.0015	1.0833 ± 0.0030	0.1976 ± 0.0017

The ratio of efficiencies between the two channels comes out to $\epsilon_{total}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)/\epsilon_{total}(B_c^+ \to J/\psi\,\pi^+) = 0.1976 \pm 0.0017$. The majority of the difference between the two channels comes from the $M(J/\psi\,\mu) > 5300$ cut used in the semileptonic channel: $\epsilon_{mass} = (17.285 \pm 0.053)\%$. We denote the efficiency within the accepted $M(J/\psi\,\mu)$ range by $\epsilon'_{total}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)$, and find that its ratio with $\epsilon_{total}(B_c^+ \to J/\psi\,\pi^+)$ is close to unity: $\epsilon'(B_c^+ \to J/\psi\,\mu^+\nu_\mu)/\epsilon(B_c^+ \to J/\psi\,\pi^+) = 1.143 \pm 0.010$.

8 Results for $\frac{\mathcal{B}(B_c^+ \to J/\psi\,\pi^+)}{\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)}$

The ratio of branching ratios R is determined from the ratio of the efficiency corrected signal yields as:

$$R = \frac{\mathcal{B}(B_c^+ \to J/\psi\,\pi^+)}{\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)} = \frac{n_{sig}(B_c^+ \to J/\psi\,\pi^+)}{n_{sig}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)} \frac{\epsilon_{total}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)}{\epsilon_{total}(B_c^+ \to J/\psi\,\pi^+)}, \tag{12}$$

giving 0.0469 ± 0.0028. The efficiency $\epsilon_{total}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)$ includes the extrapolation from $M(J/\psi\,\mu) > 5300$ MeV to the full mass range. We also quote the result for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel limited to the phase space region of $M(J/\psi\,\mu) > 5300\,\text{MeV}$ as

$$R_{M(J/\psi\,\mu)>5300} = \frac{n_{sig}(B_c^+ \to J/\psi\,\pi^+)}{n_{sig}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)} \frac{\epsilon'_{total}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)}{\epsilon_{total}(B_c^+ \to J/\psi\,\pi^+)} \tag{13}$$

which gives $R_{M(J/\psi\,\mu)>5300} = 0.271 \pm 0.016$. Systematic errors are assigned in the following sections.

9 Alternative Signal Extraction Methods

9.1 Signal Yield Extraction Via DLL Fitting

9.1.1 Fitting DLL In $B_c^+ \to J/\psi\,\mu^+\nu_\mu$

In the fit to the $M(J/\psi\,\mu)$ distributions we rely on the B background MC to constrain the background shape. While statistical uncertainty is automatically propagated to the error on the signal yield by the simultaneous fit to the data and the background MC, systematic uncertainty needs also to be probed. Therefore, we perform alternative signal extraction method which does not depend on the $M(J/\psi\,\mu)$ background shape. We fit DLL distribution instead. The upper mass limit is lowered to eliminate the upper sideband: $5300 < M(J/\psi\,\mu) < 6100\,\mathrm{MeV}$. In order to describe the signal and background DLL shapes we use an Amoroso function:

$$\mathrm{Amoroso}(x|a,\theta,\alpha,\beta) = \left(\frac{x-a}{\theta}\right)^{\alpha\beta-1} \exp\left(-\left(\frac{x-a}{\theta}\right)^{\beta}\right), \tag{14}$$

with the shape parameters constrained by the MC simulations. The fit of the model to the data was performed simultaneously to the fit of the Amoroso curves to the signal and background MC. The feeddown simulations showed that their DLL distributions are very similar to the signal shape, therefore feeddown is absorbed into the signal component. The signal yield is then corrected according to the expected feeddown fraction α. The fit results are summarized in Table 13. The fit is displayed in Figs. 24 (data) and 25 (signal and background MC). We observe $N_{J/\psi\,\mu} = 4049.7 \pm 96.8$ events. After subtracting the expected feed down contribution which is estimated in the usual manner we obtain $N_{sig} = 3967.7 \pm 94.8$ signal events.

Using the signal yield and efficiency of the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel as determined in the DLL fitting method, we calculate the ratio of branching ratios R to be

$$R = \frac{\mathcal{B}(B_c^+ \to J/\psi\,\pi^+)}{\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)} = \frac{n_{sig}(B_c^+ \to J/\psi\,\pi^+)}{n_{sig}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)} \frac{\epsilon_{total}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)}{\epsilon_{total}(B_c^+ \to J/\psi\,\pi^+)} = 0.0471 \pm 0.0025 \tag{15}$$

This is only +0.5% away from the result based on $M(J/\psi\,\mu)$ analysis. We take this as a systematic error in the background simulations.

9.1.2 Fitting DLL in $B_c^+ \to J/\psi\,\pi^+$

Even though not used in our analysis we also extracted the signal yield for the $B_c^+ \to J/\psi\,\pi^+$ channel via fitting the DLL distribution in the same manner as for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel as an additional cross-check of the DLL-fit approach. The $M(J/\psi\,\pi)$ mass range

Table 13: Amoroso fit parameters returned by simultaneous fit to DLL distributions of signal MC, background and MC, and data.

a_{sig}	-11.94 ± 0.13
θ_{sig}	$(16.97 \pm 2.05) \times 10^{-4}$
α_{sig}	49.07 ± 1.64
β_{sig}	0.4573 ± 0.0046
a_{bkg}	-11.00 ± 0.05
θ_{bkg}	20.07 ± 0.12
α_{bkg}	$0.3088 \pm .0018$
β_{bkg}	9.20 ± 0.37
N_{sig}	3967.7 ± 94.8

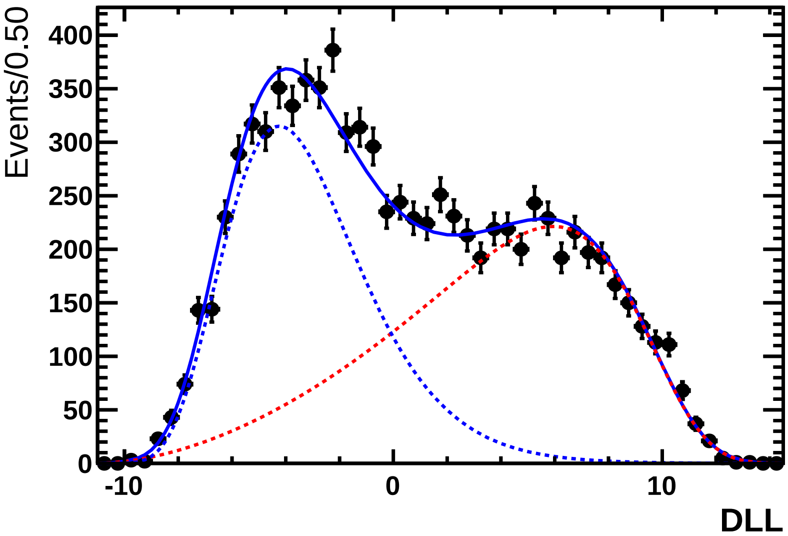


Figure 24: $B_c^+ \to J/\psi\, \mu^+ \nu_\mu$ Channel: The signal (dashed blue) and background (dashed red) contributions in the data as disentangled by the simultaneous fit to the data, signal MC and background MC.

was taken to be $m_{B_c^+} \pm 2\sigma$. As for the $B_c^+ \to J/\psi\, \mu^+ \nu_\mu$ channel, the signal DLL shape was taken from the signal MC. The background shape was obtained using the near sidebands in the data. For the left near sideband, the window was chosen as $6135.0\,\text{MeV} < M(J/\psi\pi) < 6210.0\,\text{MeV}$. At lower values of $M(J/\psi\pi)$, there is a clear presence of partially reconstructed events (see Fig. 28b) which is not representative of the background in the signal region. The upper near sideband was set to $6340\,\text{MeV} < M(J/\psi\pi) < 6490\,\text{MeV}$. The fit to the data is shown in Fig. 26.

The fit returned $N_{J/\psi\pi} = 798.5 \pm 32.6$ events. The ratio of branching fractions using DLL fitting for the signal extraction in both channels is $R = 0.0465 \pm 0.0022$, which

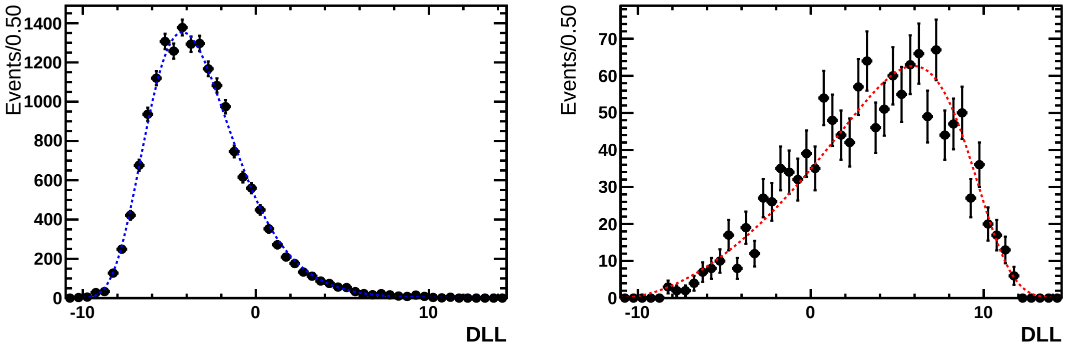


Figure 25: $B_c^+ \to J/\psi\, \mu^+ \nu_\mu$ Channel: Signal (left) and background shapes (right) superimposed on the signal MC and background MC, respectively, as determined by the simultaneous fit to the data

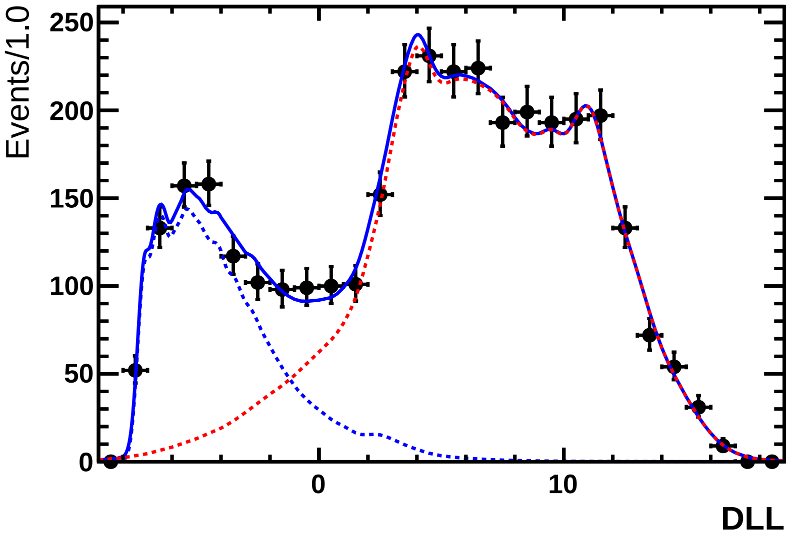


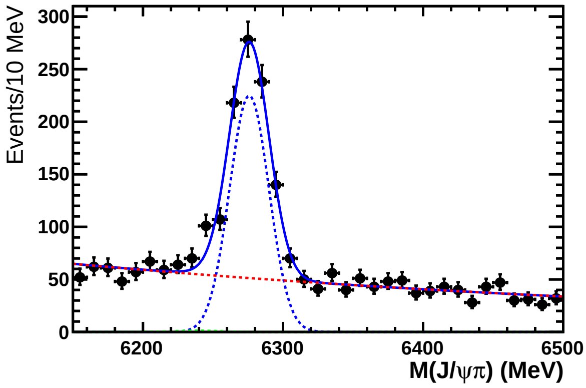
Figure 26: $B_c^+ \to J/\psi\, \pi^+$ Channel: DLL fit of signal shape (dashed blue) and background shape (dashed red) to the observed distribution in data.

is -0.8% from the nominal value. Thus we see extracting signal yield via fitting DLL distributions is an effective method.

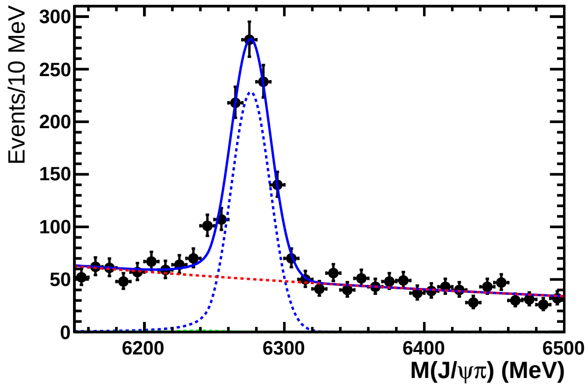
10 Systematic Errors

10.1 $M(J/\psi\, \pi)$ Signal Shape

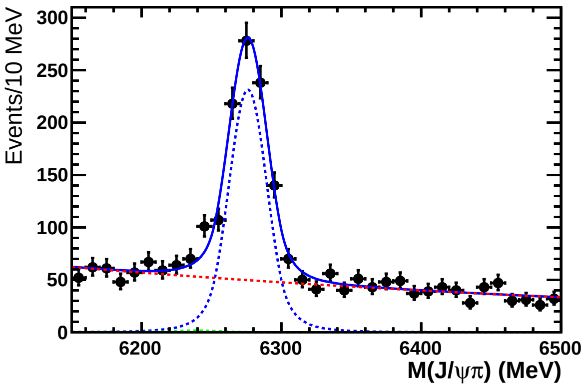
In order to study systematic errors associated with the shape of the $M(J/\psi\, \pi)$ distribution used for the signal yield extraction, different shapes were used. As an alternative to the double sided Crystal Ball distribution, the fit was also performed with a Gaussian, standard Crystal Ball, and symmetric Crystal Ball. These fits are shown in Fig. 27a, 27b, and Fig. 27c, respectively. For the different signal shapes used, the ratio of branching fractions was calculated. The results are listed in Table 14. The largest deviation from the nominal method comes from the Gaussian distribution result, for which a systematic error of 2.3% is assigned.



(a) Gaussian



(b) Single-sided Crystal Ball



(c) Symmetric double-sided Crystal Ball

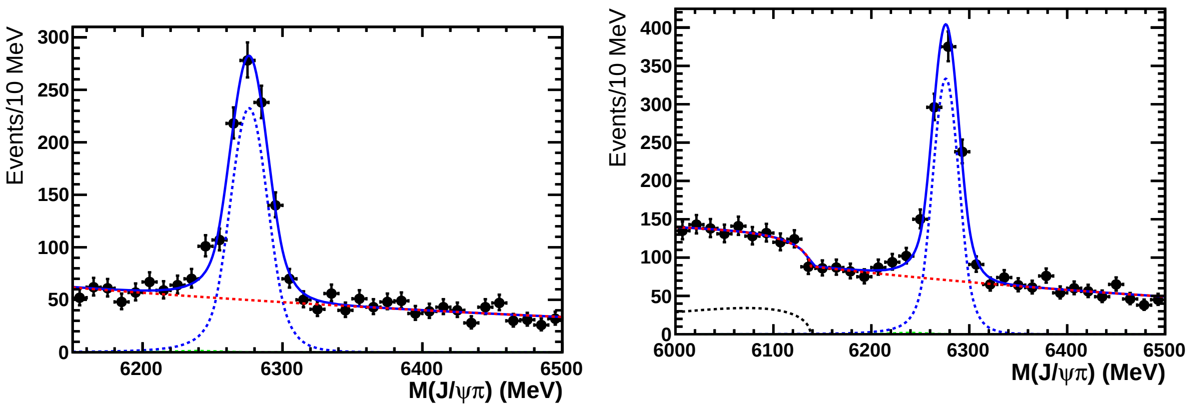
Figure 27: Various signal shape fits to $M(J/\psi\,\pi)$ distribution used in the full fit to the data.

Table 14: Summary of results for different $B_c^+ \to J/\psi\,\pi^+$ signal shapes used in the fit to $M(J/\psi\,\pi)$ distribution.

Nominal (double sided CB)	Gaussian	Standard CB	Symmetric CB
0.0469 ± 0.0028	0.0458 ± 0.0027	0.0461 ± 0.0028	0.0470 ± 0.0028
—	-2.3%	-1.6%	$+0.2\%$

10.2 $M(J/\psi\,\pi)$ Background Shape

To examine systematic errors from the background parameterization in the $M(J/\psi\,\pi)$ fit, two alternative fits are considered. The first model examined included a higher order term in the exponential, i.e. $\exp(ax) \rightarrow \exp(a_0x + a_1x^2)$, instead of simple exponential background used in the nominal fit. The fit for this background model is shown in Fig. 28a. The fit returns $n_{sig} = 838.3 \pm 40.1$. The other alternative is to widen the mass fit range to $6000.0 < M(J/\psi\,\pi) < 6500.0$. This opens up the fit to contributions from B_c decays with multiple pions and partially reconstructed tracks. To account for this, an Argus function is included as a component in the fit. This fit is shown in Fig. 28b. The fit returned $n_{sig} = 840.5 \pm 39.8$. The fit results for the default fit and the two alternative models are summarized in Table 15. We thus see that the signal yield extraction is quite stable for different background parameterizations. A systematic error of 0.2% is assigned, corresponding to the difference between the nominal and $\exp(a_0x + a_1x^2)$ shape.



(a) Background model includes higher order polynomial in exponential: $\exp(a_0x + a_1x^2)$

(b) Background is usual exponential with an additional Argus component and fit window widened.

Figure 28: Alternative background shape fits to $M(J/\psi\,\pi)$ distribution used in the full fit to the data.

Table 15: Summary of results for different $B_c^+ \rightarrow J/\psi\,\pi^+$ background shapes used in the fit to $M(J/\psi\,\pi)$ distribution.

	Default: $\exp(ax)$	$\exp(a_0x + a_1x^2)$	$\exp(ax)$+ Argus
n_{sig}	839.0 ± 40.1	838.3 ± 40.1	840.5 ± 39.8
	—	-0.1%	$+0.2\%$

10.3 $B_c \to J/\psi\, K$ Component

In the $M(J/\psi\,\pi)$ fit to the data, the contribution is fixed via the estimation from

$$\frac{N_{J/\psi\, K}}{N_{J/\psi\,\pi}} = \left(\frac{\mathcal{B}(B_c \to J/\psi\, K)}{\mathcal{B}(B_c \to J/\psi\,\pi)}\right)\left(\frac{\epsilon_{J/\psi\, K}}{\epsilon_{J/\psi\,\pi}}\right) \tag{16}$$

The measurement $\frac{\mathcal{B}(B_c\to J/\psi\, K)}{\mathcal{B}(B_c\to J/\psi\,\pi)} = 0.069 \pm 0.019 \pm 0.005$ is used. The ratio of efficiencies was determined from MC simulations of both channels, with weights from the PIDCalib package determining the $PIDK < 5$ cut efficiency. The result was $\epsilon_{J/\psi\, K}/\epsilon_{J/\psi\,\pi} = 0.1402 \pm 0.0013$. In order to assign a systematic uncertainty, the fits were repeated with these two inputs varied through their respective uncertainties. In the case of either enhancing or suppressing the $B_c \to J/\psi\, K$ component, the signal yield changes by 0.1%, which we assign as a systematic error.

10.4 Feed Down Considerations

For the decay channel $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ three different sources of possible feed down were included in the $M(J/\psi\,\mu)$ fit. The feed down yield was constrained to the estimated fraction of the signal yield. The main uncertainty in the estimation comes from the theoretical input for the B_c branching fractions. In order to estimate systematic uncertainty they were varied between the lowest and the highest theoretical prediction. The feed-down fraction varies between 1.25% and 2.76%. The fits of the signal yield were repeated with α set to these extreme values, as shown in Table 16. The signal yield changes by at most 0.6% which is taken for the systematic error.

Table 16: Summary of results for variation of the feeddown fraction

	Nominal	max	min
α	2.02%	2.76%	1.25 %
Signal yield	3536.9 ± 124.5	3524.0 ± 124.9	3556.9 ± 126.2
change in R	—	+0.4%	−0.6%

10.5 Model Dependence of Extrapolation to Full Phase Space for $B_c^+ \to J/\psi\,\mu^+\nu_\mu$

There is systematic error associated with the model dependence of extrapolation of the semileptonic rate measured in $M(J/\psi\,\mu) > 5300\,\text{MeV}$ to the full phase space. This is assigned by comparing the efficiencies of the mass cut between different MC models at the generator level. In addition to the Kiselev (nominal), Ebert, and ISGW2 models, we implemented three other models in EvtGen to be studied at the generator level. Two models [29] [23] consist of form factors calculated via the constituent light front quark model (CLF), and the other [30] based off perturbative QCD (pQCD). The invariant $M(J/\psi\,\mu)$ distributions predicted by the models at the generator level are shown in Fig. 29, and the efficiencies for the different models are listed in Table 17. The largest difference from the nominal (Kiselev) model is in the ISGW2 model, from which a systematic error of 7.9% is assigned.

We note that none of the theoretical calculations predicting form-factors is a first principle calculation (like e.g. lattice QCD). These are all phenomenological models which by definition do not have control over their theoretical uncertainties. It is not possible to favor one calculation over the other. Because of these unknown theoretical errors it is also important to sample as many theoretical calculations as possible. The spread between them is some measure of what theoretical errors are.

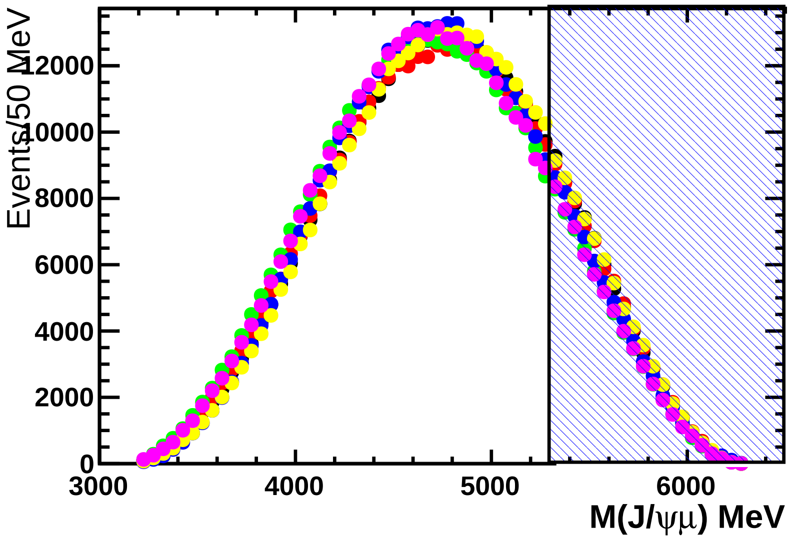


Figure 29: Invariant $M(J/\psi\,\mu)$ distributions at the generator level of the Kiselev(blue), ISGW2(yellow), Ebert(pink), WSL(black), KLL(red), WFX(green) models.

Table 17: Generator level $M(J/\psi\,\mu) > 5300\,\mathrm{MeV}$ efficiencies.

Model	ϵ_{mass}	Difference From Kiselev
Kiselev [13]	17.285 ± 0.053	—
WSL [29]	18.540 ± 0.074	+7.3%
KLL [23]	18.568 ± 0.074	+7.4%
WFX [30]	16.131 ± 0.068	−6.7%
Ebert [16]	16.068 ± 0.052	−7.0%
ISGW2 [17]	18.657 ± 0.058	+7.9%

10.6 Model Dependence of Selection Efficiencies For $B_c^+ \to J/\psi\,\mu^+\nu_\mu$

A systematic error is also assigned for the remaining model dependence of $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ selection efficiencies within the $M(J/\psi\,\mu)$ accepted range, ϵ'_{sel}. This is done by comparing the efficiency ϵ'_{sel} of the nominal (Kiselev) model with the ISGW2 and Ebert models, which differ by -1.3% and +1.3% respectively. We thus assign a systematic error of 1.3%.

10.7 Systematics of $M(J/\psi\,\mu)$ Signal Shape Used in the Fit

In order to study model dependence of the signal shape used in the $M(J/\psi\,\mu)$ fit, we replace the nominal signal MC sample (Kiselev) used in the simultaneous fit to the data and MC with the two other fully simulated models (Ebert and ISGW2). We also perform the fit without any simultaneous fit to MC. The signal yields obtained with these fit variations are listed in Table 18. The biggest difference with the nominal comes from the fit in which no model is used, for which a systematic uncertainty of 0.7% is assigned.

Table 18: Variation of signal yield obtained by the fit to the data with the signal shape parameters constrained to different MC models.

Kiselev (Nominal)	Ebert	ISGW2	No Model
3536.9 ± 124.5	3528.7 ± 125.4	3549.8 ± 125.3	3512.8 ± 127.7
—	$+0.2\%$	-0.4%	$+0.7\%$

10.8 Systematics of $M(J/\psi\,\mu)$ Background Shape used in the Fit

The functional form used to model the background $M(J/\psi\,\mu)$ distribution was varied to explore related systematic uncertainty. Alternative background functions consisted of $\exp(b_1 M(J/\psi\,\mu)) + \exp(b_2 M(J/\psi\,\mu))$ + Gaussian tail, $\exp(b_1 M(J/\psi\,\mu) + b_2 M(J/\psi\,\mu)^2)$ (no Gaussian tail), and a RooKeysPDF constructed from the background MC in the mass region of $5100 < M(J/\psi\,\mu) < 8000$. The signal yields obtained with these functions are listed in Table 19. The largest difference with the nominal result is used to assign a systematic uncertainty of 1.8%.

In all these fits the background shape is constrained by the background MC, included in simultaneous fit to the data. As discussed previously, in order to probe for possible MC mismodeling of the $M(J/\psi\,\mu)$ background shape in the signal region, we compare the result from the nominal method of fitting $M(J/\psi\,\mu)$ distribution with the results for the completely independent method of fitting DLL in the signal region. When fitting DLL, the result is 0.0471 ± 0.0025 i.e. a +0.5% difference, which we include among the systematic errors.

Table 19: Summary of signal yield fit results for different background shapes used.

$\exp(b_1 M(J/\psi\,\mu) + b_2 M(J/\psi\,\mu)^2)$ + tail(nominal)	3536.9 ± 124.5	—
$\exp(b_1 M(J/\psi\,\mu)) + \exp(b_2 M(J/\psi\,\mu))$ + tail	3555.6 ± 119.8	-0.5%
$\exp(b_1 M(J/\psi\,\mu) + b_2 M(J/\psi\,\mu)^2)$ (no tail)	3472.9 ± 126.8	$+1.8\%$
RooKeysPDF	3565.3 ± 82.0	-0.8%

10.9 Upper $M(J/\psi\,\mu)$ Fit Range Limit

The default fit range is $5300 < M(J/\psi\,\mu) < 8000$ MeV. The region above 6300 MeV contains no signal and helps determine the background level. We investigate stability of our results as a function of the upper mass limit used in the fit; i.e. we redo the fit in the range of $5300 < M(J/\psi\,\mu) < X$ for a variety of values for X. The results are shown in Table 20, where it is seen that the signal yield is quite stable. The difference between the nominal and $X = 6750$ MeV is used to assign a systematic uncertainty of 1.5%.

Table 20: Summary of signal yield fit results for different fit ranges $5300 < M(J/\psi\,\mu) < X$ MeV.

8250	8000 (nominal)	7750	7500	7250	7000	6750
3534.2 ± 123.5	3536.9 ± 124.5	3525.3 ± 126.8	3522.8 ± 128.2	3526.3 ± 128.3	3506.8 ± 133.0	3483.7 ± 145.2
$+0.1\%$	—	$+0.3\%$	$+0.4\%$	$+0.3\%$	$+0.9\%$	$+1.5\%$

10.10 Lower $M(J/\psi\,\mu)$ Fit Range Limit

In order to check stability of our results against $B \to J/\psi\,K$ background tail, we narrow the fit region to $M(J/\psi\,\mu) > 5500\,\text{MeV}$ which essentially eliminates this contribution. The fit model is kept the same, and the fit to the data can be seen in Fig. 30. The fit returns 2014.7 ± 72.8 signal events. The result for a ratio of branching fractions, $R = 0.0461 \pm 0.0028$, is -1.3% off from the nominal value. As an additional check we also extended the mass window to lower value, $M(J/\psi\,\mu) > 5100\,\text{MeV}$, which makes the $B \to J/\psi\,K$ background component large, however still eliminates decays with $B \to J/\psi +$ multiple hadrons. The fit is shown in Fig. 31. The fit returns 5549.1 ± 222.2 events. The result for a ratio of branching fractions, $R = 0.0463 \pm 0.0029$, is -1.6% off from the nominal value.

We conclude, from these studies that the results are stable with respect to the $M(J/\psi\,\mu)$ cut-off and assign a 1.6% systematic error.

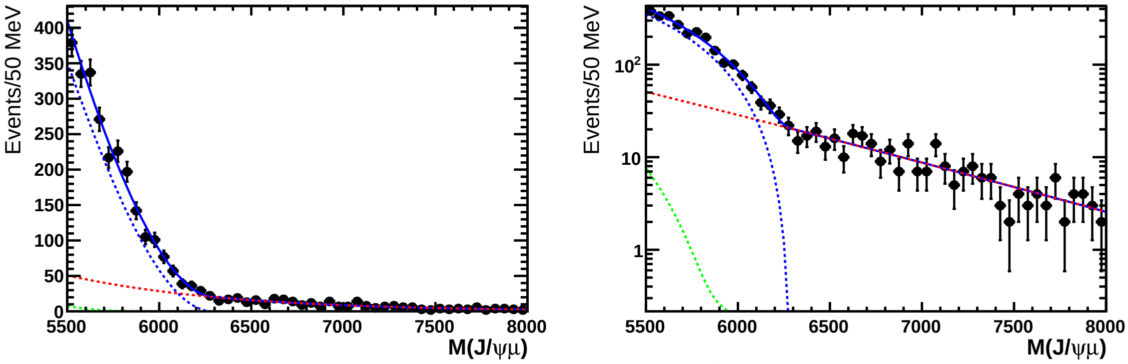


Figure 30: Full fit to the data for narrowed window of $M(J/\psi\,\mu) > 5500\,\text{MeV}$ with signal (dashed blue), background (dashed red), and feed down(dashed green) components shown.

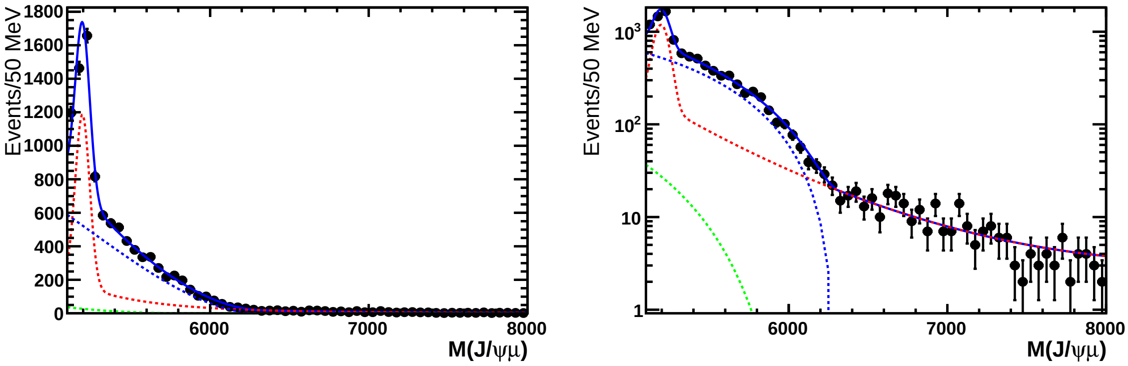


Figure 31: n Full fit to the data for widened window of $M(J/\psi\,\mu) > 5100\,\text{MeV}$ with signal (dashed blue), background (dashed red), and feed down(dashed green) components shown.

10.11 PID Systematics

There is a systematic error associated with the PID cuts on the π^+ in the $B_c^+ \to J/\psi\,\pi^+$ analysis and the bachelor μ^+ in the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ analysis. For the $PIDK < 5$ cut we use the PID Calib package for determining efficiencies from the data. The efficiencies are done in bins for momentum, rapidity, and track multiplicity. The main source of systematic uncertainty then comes from the assumption by the calibration method that PID cut efficiencies do not vary within the bins used for calibration. To assign the systematic error then, a single bin is used for each variable binned in. This results in a 0.8% efficiency difference, for which the systematic uncertainty is assigned.

For the bachelor muon, there is both an IsMuon cut in the stripping and the $PIDmu > 0$. To determine the systematic error associated with the IsMuon cut, values from the Performance of the Muon Identification at LHCb [31] paper were used. From this, a 1.5% systematic error is associated with the average IsMuon efficiency loss for $p_T < 5\,\text{GeV}$. Also, the $PIDmu < 0$ cut was removed and the recalculated ratio of branching fractions was 0.0471 ± 0.0029, a +0.4% difference. Thus a total systematic error of 1.9% is assigned for the muon PID cuts.

10.12 Lifetime Systematics

In order to study systematic uncertainties associated with the lifetime, the MC was weighted to different values. Each event was given a weight according to $\exp(-\tau_d/\tau_n)/\exp(-\tau_d/\tau_{MC})$ where τ_d is the proper decay time of the particular event extracted from the truth information, τ_n is the new lifetime to which the MC is being reweighted, and τ_{MC} is the lifetime the MC was generated with. Both $B_c^+ \to J/\psi\,\pi^+$ and $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ MC samples were reweighted this way and efficiencies were recalculated. As is seen in Fig. 32, there is only a weak linear dependence of R on the particular lifetime used. A systematic error of 0.2% is assigned based off of the result when reweighting to the errors on the world average: 500 ± 13 fs.

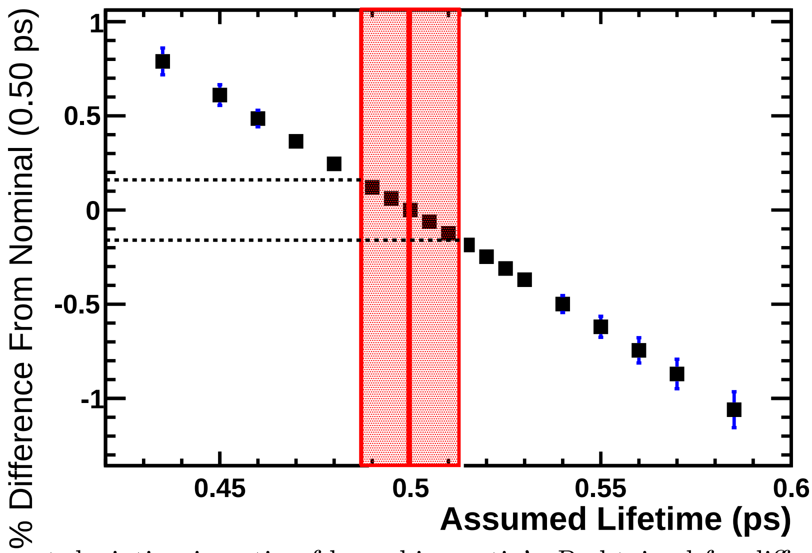


Figure 32: Percent deviation in ratio of branching ratio's R obtained for different $\tau(B_c)$ values used in the MC. The red band represents the world average and its errors.

10.13 Multiple Candidates

For the $B_c^+ \to J/\psi\,\pi^+$ channel 0.12% of events in the fit region are multiple candidate events, and for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel, 1.87% of events. In order to study the effects a random kill procedure was used, in which one of the candidates in the multiple candidate events was randomly chosen to be cut and the fits redone. The effect is completely negligible in the hadronic channel. In the semileptonic channel, the signal yield resulting from fitting the data dropped by 1.92%. The random kill procedure applied to the MC, however, seems to accurately predict the behavior in the data. Therefore, results in the end ratio of branching fractions change by only +0.4%, for which a systematic error is assigned.

10.14 MC/Data DLL Differences

We note that in the semileptonic channel, we already include in the systematic errors a direct fit of the DLL distributions which covers this. In order to probe agreement in the $B_c^+ \to J/\psi\pi^+$ channel we redo fits for various DLL cuts. We look for changes relative to the nominal cut: $DLL < 1$. In particular we look at the percent difference as:

$$\left(\frac{y_{data}}{y_{mc}} - 1\right) \times 100 \tag{17}$$

for

$$y = \frac{n_{sig}(DLL < X)}{n_{sig}(DLL < 1)} \tag{18}$$

The results are shown in Fig33. In general deviations are small, and come with large errors. Based off this study we will assign a systematic uncertainty of 2%

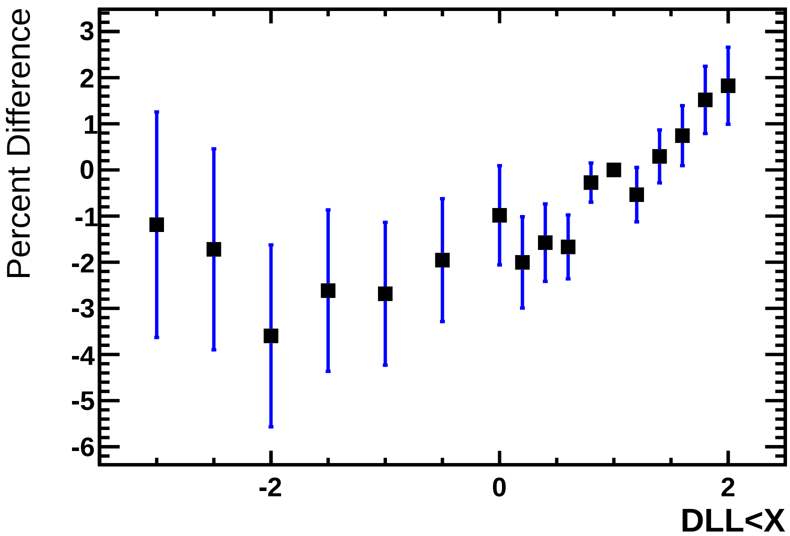


Figure 33: Variation in $y = \frac{n_{sig}(DLL<X)}{n_{sig}(DLL<1)}$ for the $B_c^+ \to J/\psi\pi^+$ channel resulting from varying DLL cut from the nominal ($DLL < 1$).

10.15 Trigger Systematics

To assess systematic uncertainties associated with the trigger, we examine the stability of our results upon removing different lines in turn from the trigger requirement. The results are shown in Table 21. Overall there is good agreement in the Hlt lines, but some discrepancy in L0.

It was suggested that perhaps more consistent results could be obtained by requiring TOS on J/ψ, with the reason being that it should be more symmetric choice between the

Table 21: Changes upon removing individual lines from the trigger requirements. The ϵ columns show the drop in efficiency as determined from the MC when a particular line is removed. The last column shows how the ratio of branching fractions R changes.

Line Removed	$\epsilon_{J/\psi\pi}$ drop	$\epsilon_{J/\psi\mu}$ drop	% Deviation of R
(1) BcL0MuonDecision$_{\mathrm{TOS}}$	0.7137 ± 0.0011	0.9131 ± 0.0023	-2.1344 ± 2.1272
(2) BcL0DiMuonDecision$_{\mathrm{TOS}}$	0.9976 ± 0.0001	0.9972 ± 0.0004	$+3.3554 \pm 0.8633$
(3) BcHlt1TrackAllL0Decision$_{\mathrm{TOS}}$	0.9537 ± 0.0005	0.9967 ± 0.0005	-0.2712 ± 0.8527
(4) BcHlt1TrackMuonDecision$_{\mathrm{TOS}}$	0.9861 ± 0.0003	0.9875 ± 0.0009	$+0.6409 \pm 0.5642$
(5) BcHlt1DiMuonHighMassDecision$_{\mathrm{TOS}}$	0.9244 ± 0.0006	0.9223 ± 0.0022	-1.0481 ± 1.4134
(6) BcHlt2DiMuonDetachedJPsiDecision$_{\mathrm{TOS}}$	0.9515 ± 0.0005	0.9753 ± 0.0012	$+1.1902 \pm 0.6737$
(7) BcHlt2DiMuonDetachedHeavyDecision$_{\mathrm{TOS}}$	0.9998 ± 0.0	0.9882 ± 0.0009	$+0.7279 \pm 0.2826$
(8) BcHlt2TopoMu2BodyBBDTDecision$_{\mathrm{TOS}}$	0.965 ± 0.0005	0.9781 ± 0.0012	-0.4043 ± 0.7508
(9) BcHlt2TopoMu3BodyBBDTDecision$_{\mathrm{TOS}}$	0.9999 ± 0.0	1.0 ± 0.0	$+0.0117 \pm 0.0157$

two channels. Thus the study was repeated for TOS on J/ψ. The results are shown in Table 22

Table 22: The same as table 21, but with requiring TOS on J/ψ.

Line Removed	$\epsilon_{J/\psi\pi}$ drop	$\epsilon_{J/\psi\mu}$ drop	% Deviation of R
(1) BcL0MuonDecision$_{\mathrm{TOS}}$	0.7234 ± 0.0011	0.7136 ± 0.004	-5.832 ± 2.3622
(2) BcL0DiMuonDecision$_{\mathrm{TOS}}$	0.9976 ± 0.0001	0.9973 ± 0.0005	$+2.1937 \pm 0.8453$
(3) BcHlt1TrackAllL0Decision$_{\mathrm{TOS}}$	0.9877 ± 0.0003	0.9894 ± 0.0009	$+0.3957 \pm 0.5405$
(4) BcHlt1TrackMuonDecision$_{\mathrm{TOS}}$	0.9741 ± 0.0004	0.97 ± 0.0015	-0.0709 ± 0.7532
(5) BcHlt1DiMuonHighMassDecision$_{\mathrm{TOS}}$	0.8722 ± 0.0008	0.8762 ± 0.0029	-2.3164 ± 1.8861
(6) BcHlt2DiMuonDetachedJPsiDecision$_{\mathrm{TOS}}$	0.9423 ± 0.0006	0.9572 ± 0.0018	$+2.2012 \pm 0.9659$
(7) BcHlt2DiMuonDetachedHeavyDecision$_{\mathrm{TOS}}$	1.0 ± 0.0	1.0 ± 0.0	$+0.0006 \pm 0.0006$
(8) BcHlt2TopoMu2BodyBBDTDecision$_{\mathrm{TOS}}$	1.0 ± 0.0	1.0 ± 0.0	$+0.0496 \pm 0.0416$

Thus, we see that we get less consistent results using J/ψ TOS than when we require TOS on B_c. We also note that the ratios of efficiencies between the different TOS choices for the hadronic channel is

$$\epsilon(J/\psi\pi)_{TOS(J/\psi)}/\epsilon(J/\psi\pi)_{TOS(B_c)} = 0.93457 \pm 0.00061,$$

whereas a bigger drop is observed in the semileptonic channel;

$$\epsilon(J/\psi\mu)_{TOS(J/\psi)}/\epsilon(J/\psi\mu)_{TOS(B_c)} = 0.8459 \pm 0.0029.$$

Furthermore, we have

$$[N(J/\psi\pi)/\epsilon(J/\psi\pi)]_{TOS(J/\psi)}/[N(J/\psi\pi)/\epsilon(J/\psi\pi)]_{TOS(B_c)} = 1.0076 \pm 0.0087,$$

which means that the data follow the trigger simulations in $B_c^+ \to J/\psi\,\pi^+$ channels. However, the similar quantity for $B_c^+ \to J/\psi\,\mu^+\nu_\mu$,

$$[N(J/\psi\mu)/\epsilon(J/\psi\mu)]_{TOS(J/\psi)}/[N(J/\psi\mu)/\epsilon(J/\psi\mu)]_{TOS(B_c)} = 0.9455 \pm 0.0085$$

deviates from unity implying differences between the data and the simulations.

Since, the semileptonic channel is a three-body decay it leads to a softer J/ψ than in $B_c^+ \to J/\psi\,\pi^+$ decay. Therefore, narrowing TOS to J/ψ is not necessarily expected to lead to a decrease in trigger systematics. In fact, from the evidence presented above we conclude that the better strategy is to require TOS on B_c which maximizes trigger efficiency, especially in poorly simulated semileptonic channel, and therefore leaves less room for a systematic uncertainty.

Based off the results of Table 21 we assign a 3.4% systematic error corresponding to the largest deviation seen from our nominal result.

10.16 Summary of Systematic Errors

A summary of all of the sources of systematic errors are given in Table 23.

Table 23: Summary of Systematic Errors

$M(J/\psi\,\pi)$ Signal Shape	2.3%
$M(J/\psi\,\pi)$ Background Shape	0.2%
$B_c \to J/\psi\,K$ Component	0.1%
Pi PID	0.8%
$M(J/\psi\,\mu)$ Signal Shape	0.7%
Lower $M(J/\psi\,\mu)$ Fit Range Limit	1.6%
Upper $M(J/\psi\,\mu)$ Fit Range Limit	1.5%
$M(J/\psi\,\mu)$ Background Shape	1.8%
Replace $M(J/\psi\,\mu)$ fit with DLL fit	0.5%
Feed down	0.6%
$\epsilon'_{sel}(B_c^+ \to J/\psi\,\mu^+\nu_\mu)$ Model Dependence	1.3%
Muon PID	1.9%
Lifetime	0.2%
Multiple Candidates	0.4%
MC/Data DLL Agreement	2.0%
Trigger	3.4%
Total within selected $M(J/\psi\,\mu)$	6.0%
$M(J/\psi\,\mu)$ extrapolation	7.9%
Total	9.9%

11 Summary

Together with the total systematic error given in Table 23, the ratio of branching fractions is measured to be

$$R = \frac{\mathcal{B}(B_c^+ \to J/\psi\, \pi^+)}{\mathcal{B}(B_c^+ \to J/\psi\, \mu^+ \nu_\mu)} = 0.0469 \pm 0.0028(\mathrm{stat}) \pm 0.0046(\mathrm{syst}) \tag{19}$$

We also present the result for the ratio with $B_c^+ \to J/\psi\, \mu^+\nu_\mu$ limited to area of phase space with $M(J/\psi\, \mu) > 5300\,\mathrm{MeV}$. This results does not depend on model dependent extrapolation to full phase space. We quote it since it has a lower relative systematic error.

$$R_{M(J/\psi\, \mu)>5300} = 0.271 \pm 0.016 \pm 0.016(\mathrm{syst}) \tag{20}$$

There are a wide range of theoretical predictions available to compare our result. They are summarized in Table 24. Overall it is seen that the observed result R is lower than theoretical predictions, but is consistent with the calculations of Ebert, Faustov, Galkin (2003) [16] and El-Hady, Munoz, Vary(1999) [32], and Ke, Li, Liu (2013) [23]. Five other predictions are inconsistent with our result. The comparison to theoretical predictions is visualized in Fig. 34.

Table 24: Summary of theoretical predictions for R

Chang [20]	Anisimov [28]	El-Hady [32]	Colangelo [21]	Kiselev [13]	Ebert [16]	Ivanov [25]	Ke [23]
0.0909	0.0625	0.0525	0.0867	0.0684	0.0496	0.0821	0.0577

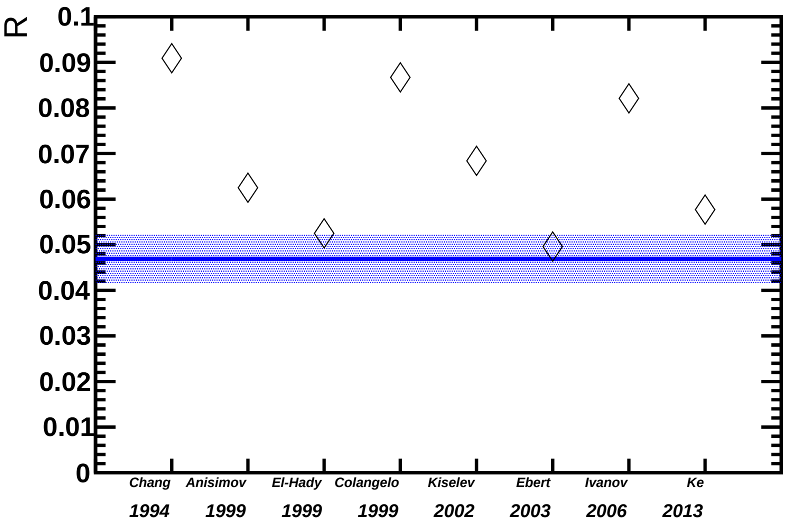


Figure 34: Comparison of our measured value of R with various theoretical predictions.

With the ratio of branching fractions R measured, we can now calculate the scaling factor of $\sigma(B_c)/\sigma(B)$ between LHCb and CDF:

$$\frac{\frac{\sigma(B_c)}{\sigma(B)}(LHCb)}{\frac{\sigma(B_c)}{\sigma(B)}(CDF)} = \frac{\frac{\sigma(B_c)\mathcal{B}(B_c^+\to J/\psi\,\pi^+)}{\sigma(B)\mathcal{B}(B^+\to J/\psi\,K)}(LHCb)}{\frac{\sigma(B_c)\mathcal{B}(B_c^+\to J/\psi\,\mu^+\nu_\mu)}{\sigma(B)\mathcal{B}(B^+\to J/\psi\,K)}(CDF)\times R} = 0.49^{+0.19}_{-0.22}$$

According to private communications with Alexandr Berezhnoy, theory predicts that this number should be ~ 1 since the leading dependence on collision energy and rapidity range (p_{T} cut-off was the same in both measurements) cancels between B_c and B cross-sections. The result above is 2.7σ below this expectation. Nevertheless, the LHCb and CDF results agree within a factor of 2, thus the observed deviation is insufficient to bridge an order of magnitude discrepancy between the expected and measured $\sigma(B_c)/\sigma(B)$ values, discussed in the introduction.

12 Appendix

12.1 MC Validation

As mentioned in Sec 4.1, the efficiency for a range of cuts on the individual variables used in the log-likelihood formalism from the MC and data were compared. For example, the behavior of the data and MC under cuts such as $\chi^2_{IP}(B_c) < X$ for different X was studied. The efficiency here is defined as a relative efficiency to the efficiency after the preselection cuts (i.e. efficiency after preselection is 100% by definition). It was observed that the resolution of the MC was too optimistic for the $\chi^2_{vtx}/NDOF(B_c)$ and $\chi^2_{IP}(B_c)$ variables. In order to bring the efficiencies of the MC into better agreement with the data, smearing was applied to the variables. The smearing was done according to Eq. 21.

$$(\chi^2_{Vtx}/NDOF(B_c))_{\text{smeared}} = \chi^2_{Vtx}/NDOF(B_c) + (F \cdot \text{rannor})^2 \tag{21}$$

The efficiency plots were made for a variety of choices of F, and the particular value was settled on by seeking to minimize the quantity use as a measure

$$M = \sum_i \left(\epsilon_i(data)(\epsilon_i(data) - \epsilon_i(MC))\right)^2 \tag{22}$$

where ϵ_i refers to the efficiency of either the data or MC for each bin. The result for M is shown in the table below for each of the smeared variables for the different choices of F.

	$B_c \to J/\psi\pi$		$B_c \to J/\psi\mu\nu$	
F	$\chi^2_{IP}(B_c)$	$\chi^2_{Vtx}(B_c)/DOF$	$\chi^2_{IP}(B_c)$	$\chi^2_{Vtx}(B_c)/DOF$
0	0.01486	0.00267	0.00317	0.00182
0.10	0.00814	0.00247	0.00134	0.00154
0.20	0.00403	0.00204	0.00201	0.00138
0.25	0.00286	0.00173	0.00116	0.00144
0.30	0.00213	0.00146	0.00104	0.00130
0.35	0.00205	0.00134	0.00075	0.00117
0.40	0.00243	0.00137	0.00114	0.00119
0.45	0.00330	0.00175	0.00102	0.00133
0.50	0.00477	0.00253	0.00126	0.00216

Thus the choice of $F = 0.35$ is best for all cases. We note that the results of the analysis do not require a precise determination of the best F. For example, use of $F = 0.30$ changes our result for ratio of branching fractions by 0.0026%, a completely negligible effect. The efficiency plots for each variable in the $B_c^+ \to J/\psi\,\pi^+$ channel are shown in Fig. 35- 38, and for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel in Fig. 39- 42.

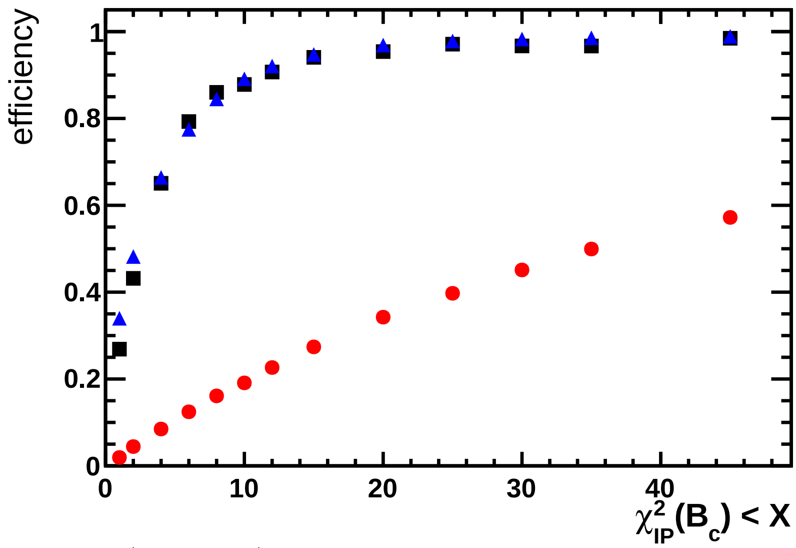


Figure 35: Fraction of $B_c^+ \to J/\psi\,\pi^+$ candidates after the preselection passing a cut of $\chi^2_{IP}(B_c) < X$, where X is given on the horizontal axis, for smeared Monte Carlo signal (blue triangles), signal in the data (black squares) and background events in the data from the far-sidebands (red circles).

12.1.1 Pseudo-rapidity and $p_{\rm T}$ distributions

Shown in Fig. 43- 46 are the comparisons for η (pseudo-rapidity) and $p_{\rm T}$ between the $B_c^+ \to J/\psi\,\pi^+$ data and the signal MC. The distributions in the data were obtained using the *sWeight* formalism. The number of MC events has been normalized to the number of signal events. The agreement between data and MC was seen to be quite good.

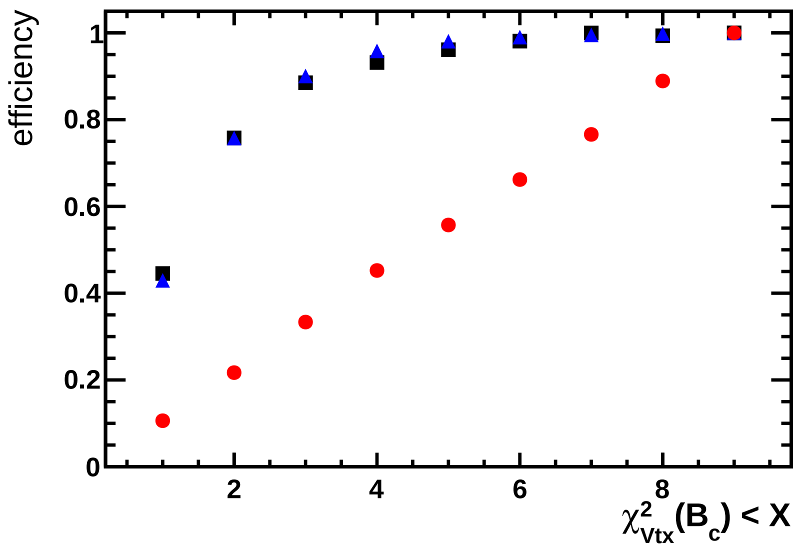


Figure 36: Fraction of $B_c^+ \rightarrow J/\psi\pi^+$ candidates after the preselection passing a cut of $\chi^2_{Vtx}/NDOF(B_c) < X$, where X is given on the horizontal axis, for smeared Monte Carlo signal (blue triangles), signal in the data (black squares) and background events in the data from the sidebands (red circles).

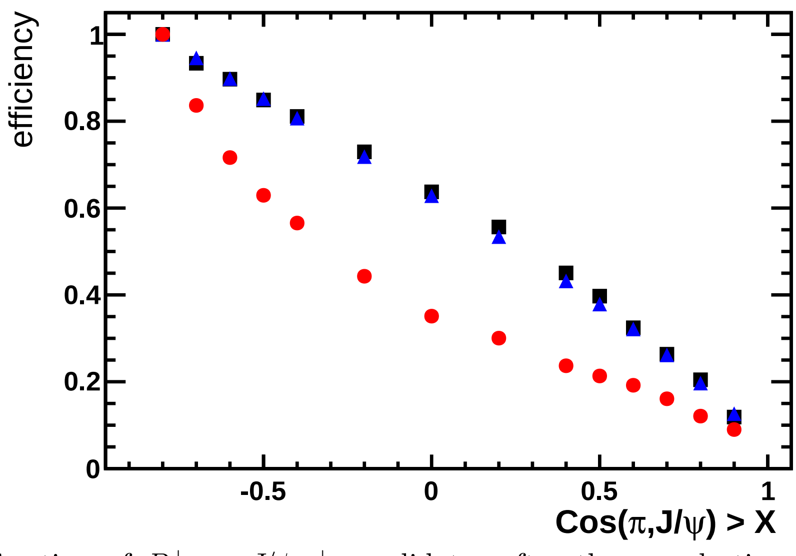


Figure 37: Fraction of $B_c^+ \rightarrow J/\psi\pi^+$ candidates after the preselection passing a cut of $\cos(\pi, J/\psi) > X$, where X is given on the horizontal axis, for Monte Carlo signal (blue triangles), signal in the data (black squares) and background events in the data from the sidebands (red circles).

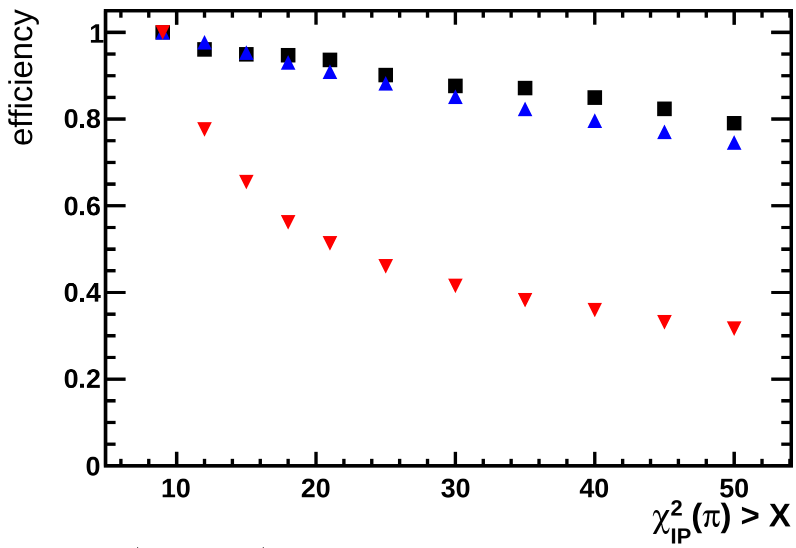


Figure 38: Fraction of $B_c^+ \to J/\psi\,\pi^+$ candidates after the preselection passing a cut of $\chi^2_{IP}(\pi) > X$, where X is given on the horizontal axis, for signal Monte Carlo (blue triangles), signal in the data (black squares) and background events in the data from the sidebands (red circles).

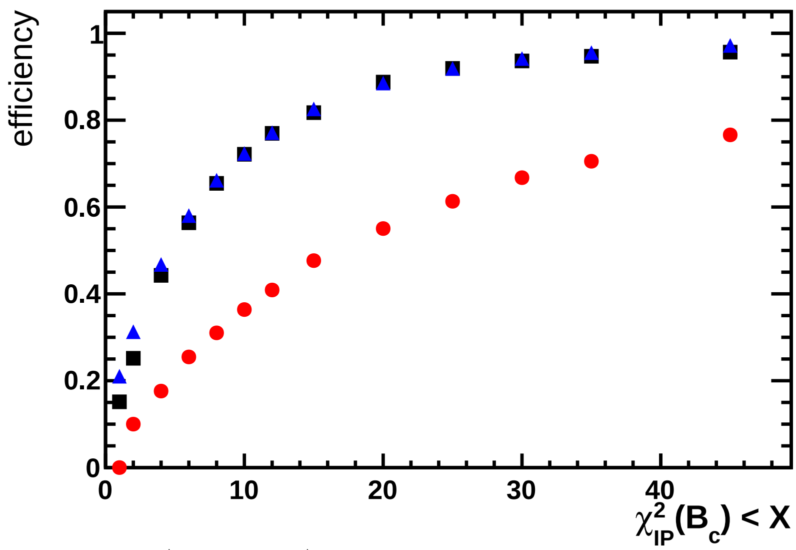


Figure 39: Fraction of $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ candidates after the preselection passing a cut of $\chi^2_{IP}(B_c) < X$, where X is given on the horizontal axis, for smeared Monte Carlo signal (blue triangles), signal in the data (black squares) and background events in the data from the far-sidebands (red circles).

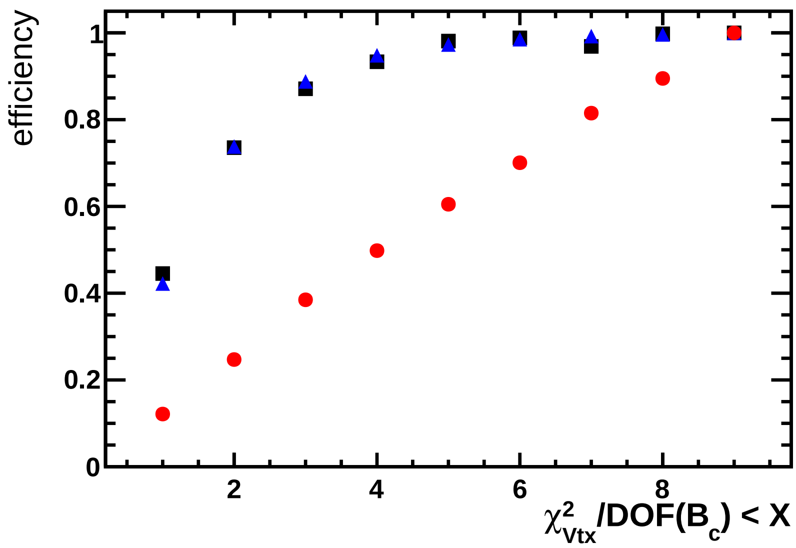


Figure 40: Fraction of $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ candidates after the preselection passing a cut of $\chi^2_{Vtx}/NDOF(B_c) < X$, where X is given on the horizontal axis, for smeared Monte Carlo signal (blue triangles), signal in the data (black squares) and background events in the data from the sidebands (red circles).

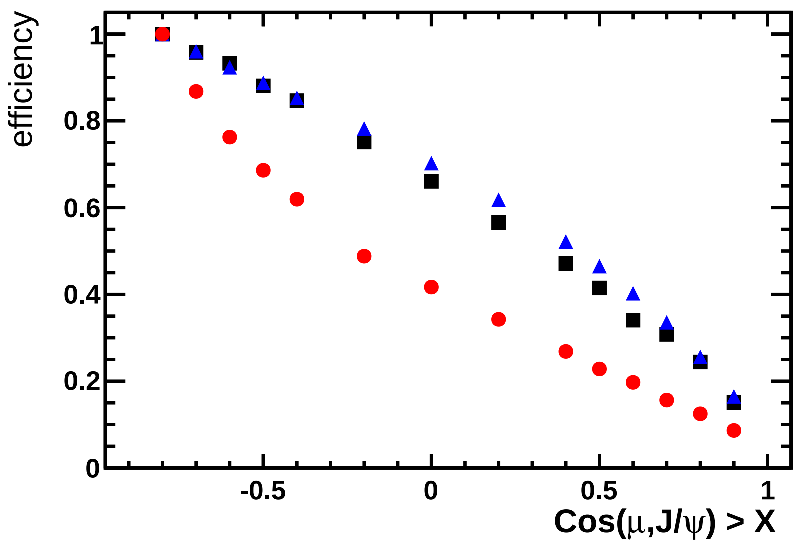


Figure 41: Fraction of $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ candidates after the preselection passing a cut of $\cos(\pi, J/\psi) > X$, where X is given on the horizontal axis, for Monte Carlo signal (blue triangles), signal in the data (black squares) and background events in the data from the sidebands (red circles).

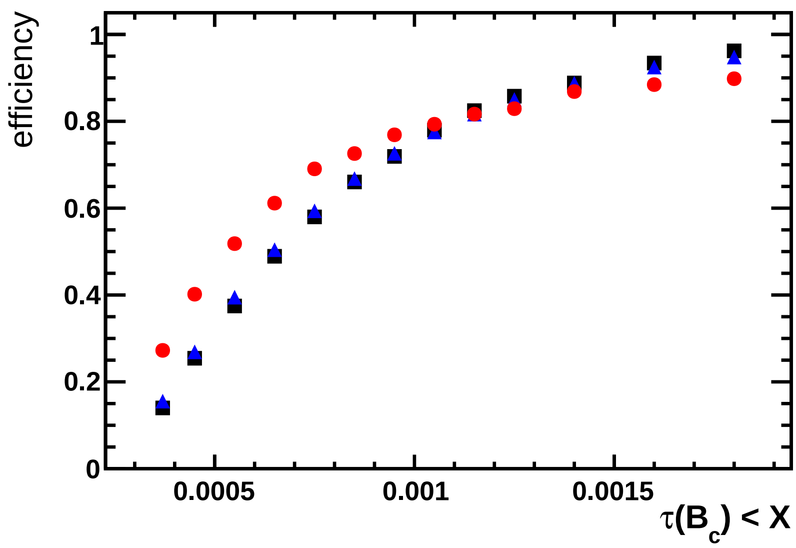
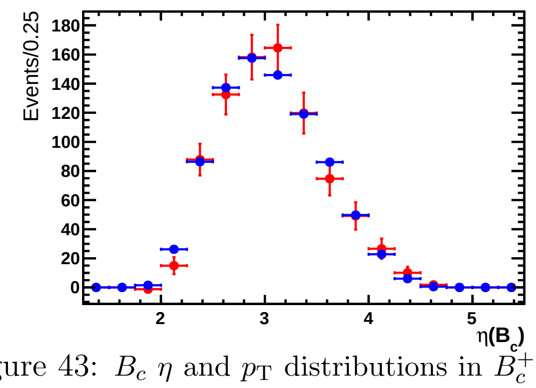


Figure 42: Fraction of $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ candidates after the preselection passing a cut of $\tau(B_c) < X$, where X is given on the horizontal axis, for signal Monte Carlo (blue triangles), signal in the data (black squares) and background events in the data from the sidebands (red circles).



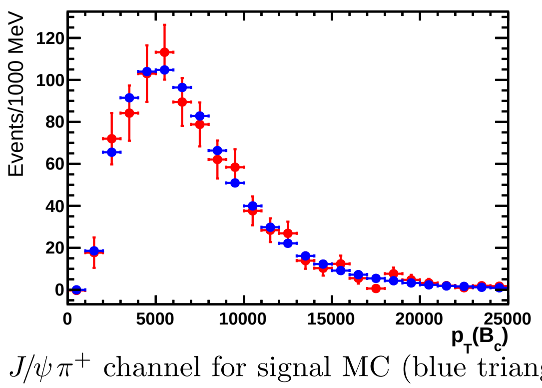


Figure 43: B_c η and p_{T} distributions in $B_c^+ \to J/\psi\,\pi^+$ channel for signal MC (blue triangles) and data using *sWeight* (red squares).

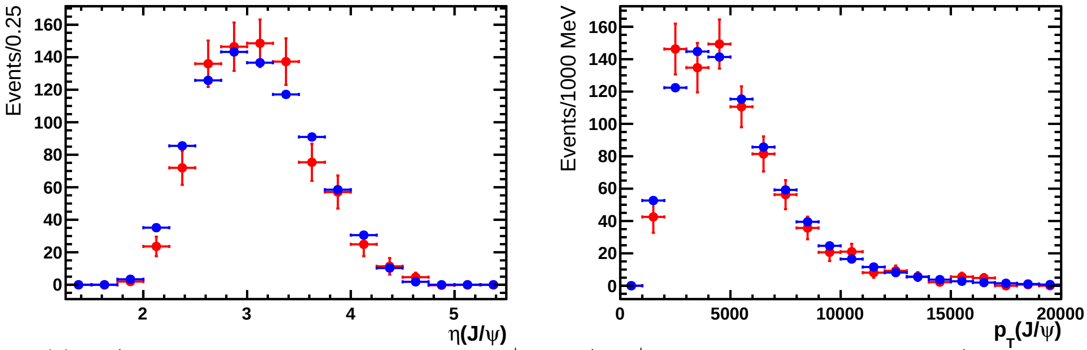


Figure 44: J/ψ η and $p_{\rm T}$ distributions in $B_c^+ \to J/\psi\pi^+$ channel for signal MC (blue triangles) and data using *sWeight* (red squares).

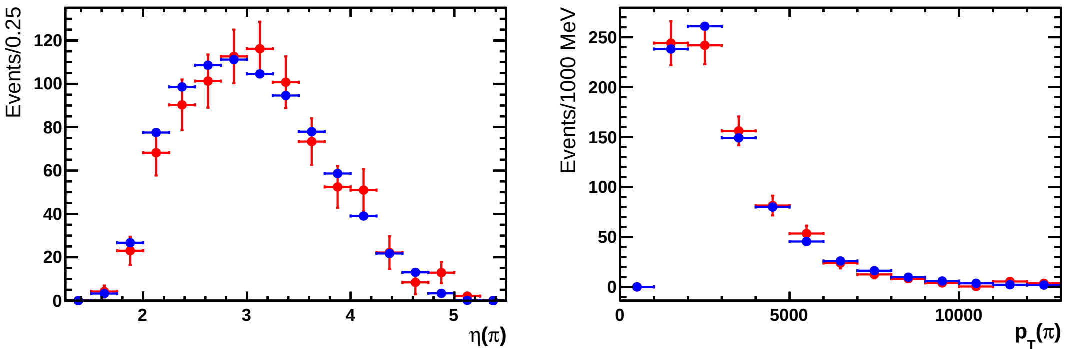


Figure 45: π η and $p_{\rm T}$ distributions in $B_c^+ \to J/\psi\pi^+$ channel for signal MC (blue triangles) and data using *sWeight* (red squares).

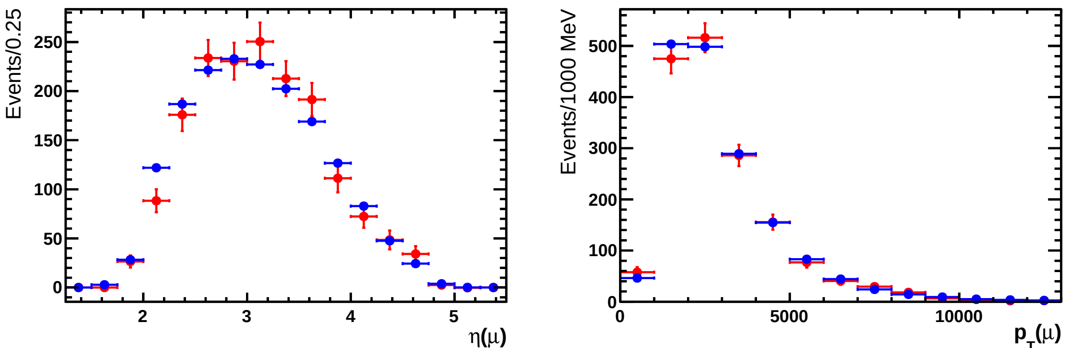


Figure 46: μ η and $p_{\rm T}$ distributions in $B_c^+ \to J/\psi\pi^+$ channel from J/ψ for signal MC (blue triangles) and data using *sWeight* (red squares).

The same comparisons were repeated for the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel. Shown in Fig. 47-50 are the comparisons for η and p_{T} between the background-subtracted $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ data and the MC. The background subtraction is performed using the distributions found in the background MC. The number of MC events has been normalized to the number of signal events. The agreement between data and MC was again seen to be quite good in this channel.

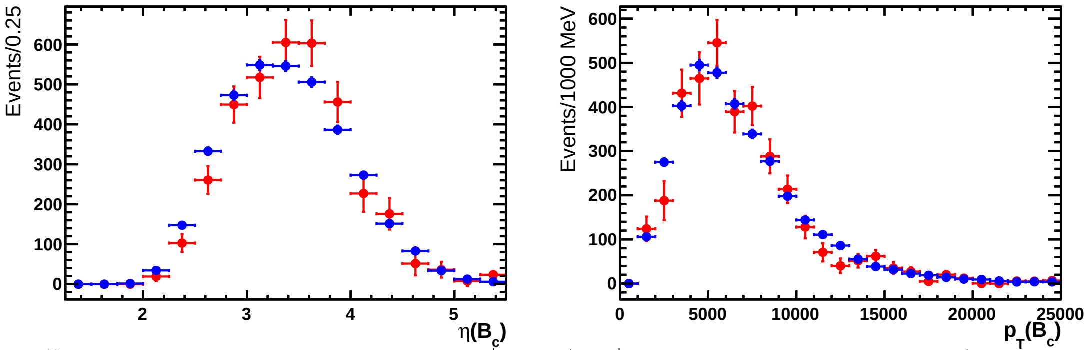


Figure 47: B_c η and p_{T} distributions in $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel for signal MC (blue triangles) and data using *sWeight* (red squares).

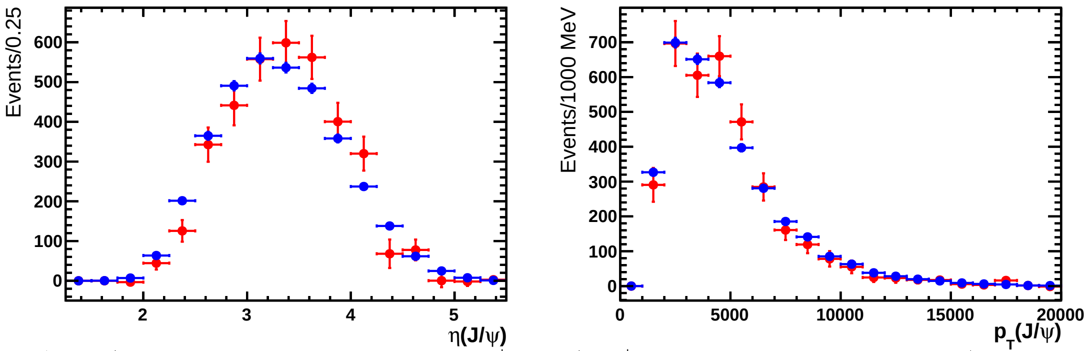


Figure 48: J/ψ η and p_{T} distributions in $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel for signal MC (blue triangles) and data using *sWeight* (red squares).

12.2 $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ q^2 Distribution

Studies were also done on the q^2 distribution of the $B_c^+ \to J/\psi\,\mu^+\nu_\mu$ channel. The q^2 distributions were obtained by first finding the neutrino momentum by solving the usual quadratic equation obtained from conservation of energy and momentum:

$$a p_\nu^{\parallel 2} + b p_\nu^{\parallel} + c = 0 \tag{23}$$

with

$$a = 4\left(m_{\mu\psi}^2 + p_{\mu\psi}^{\perp 2}\right) \qquad b = 4p_{\mu\psi}^{\parallel 2}\left(2p_{\mu\psi}^{\perp 2} - (m_{B_c}^2 - m_{\mu\psi}^2)\right) \qquad c = 4p_{\mu\psi}^{\perp 2}\left(p_{\mu\psi}^{\parallel 2} + m_{B_c}^2\right) - (m_{B_c}^2 - m_{\mu\psi}^2)^2 \tag{24}$$

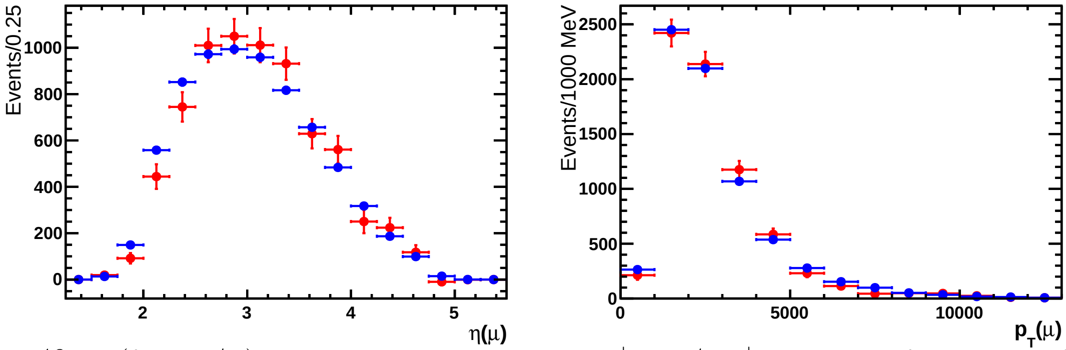


Figure 49: μ (from J/ψ) η and p_{T} distributions in $B_c^+ \to J/\psi\, \mu^+ \nu_\mu$ channel for signal MC (blue triangles) and data using *sWeight* (red squares).

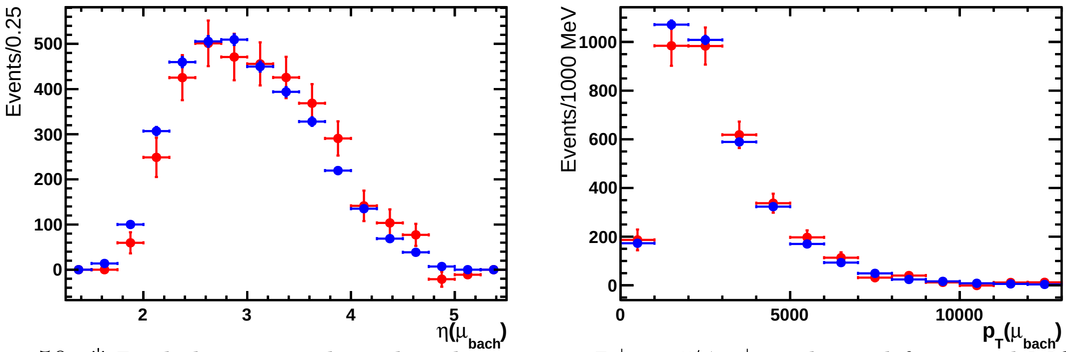


Figure 50: * Bachelor μ η and p_{T} distributions in $B_c^+ \to J/\psi\, \mu^+ \nu_\mu$ channel for signal MC (blue triangles) and data using *sWeight* (red squares).

where the $\perp$ and $\|$ symbols mark the components of the momentum perpendicular and parallel to the B_c flight path direction obtained from the vertex information, and m_{B_c} is taken at the PDG value [8]. There are two possible solutions for the neutrino momentum. For each event, they are classified according to if the neutrino makes a larger or smaller angle with the B_c flight path. There is also the issue of the quadratic equation resulting in a negative discriminant. In the selected region of phase space the neutrino is relatively soft, which results in resolution effects causing a negative discriminant more frequently. In these cases, the event is thrown out. This results in only $\sim 40\%$ of signal events surviving. The different solutions are shown in Fig 51, along with the q^2 distributions of the Kiselev (nominal), Ebert, and ISGW2 models. There is little difference between the three MC models, and they all agree reasonably well with the data. The q^2 resolution in the endpoint region is poor, dilluting sensitivity of this variable to the model differences.

12.3 Investigation of the pole mass in the Kiselev model

To extrapolate the measured $B_c^+ \to J/\psi\, \mu^+ \nu_\mu$ endpoint rate to the full phase-space we use the Kiselev's model for the central value of the extrapolation factor, and rely on the other models for an estimate of the systematic error. In this additional study, we vary one of

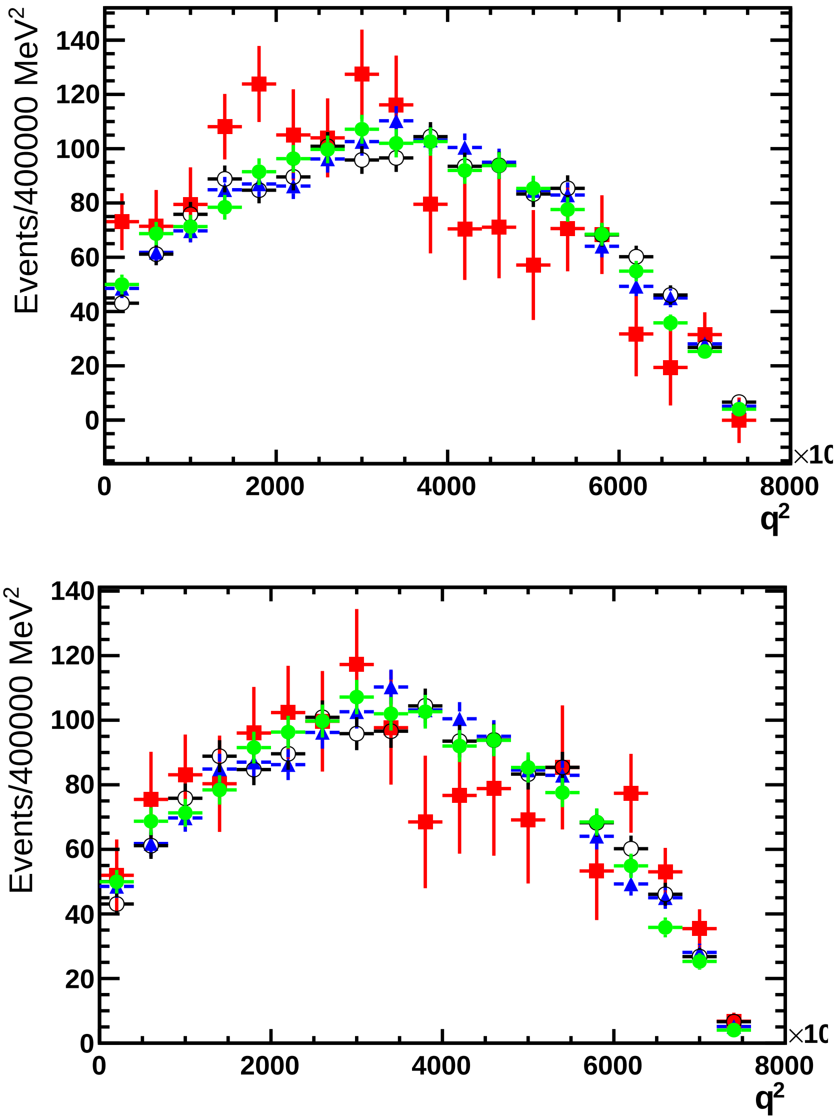


Figure 51: q^2 distributions of the data (red squares) and MC models: Kiselev (blue triangles), Ebert (open black circles), ISGW2 (filled green circles). The top plot and bottom plots correspond to the q^2 value obtained when the neutrino momentum solution is taken that results in a smaller and larger angle between the neutrino and with the B_c flight path, respectively.

the key parameters of the Kiselev's model, to see if its value can be well constrained from the data themselves, and therefore minimize the extrapolation error.

In the Kiselev model the form factors are evaluated at $q^2 = 0$ and evolved to the whole physical region using the pole dependence

$$F_i(q^2) = \frac{F_i(0)}{1 - q^2/M_i^2} \tag{25}$$

where the subscript i refers to the particular form factor. The nominal value used for the pole mass is $M_i = 4500\,\mathrm{MeV}$ for all four form factors. Generator level MC events were produced for variations of the pole mass M_i value in a wide range from 3250 MeV to infinity (the maximum q^2 and thus minimum M_i^2 is $(m_{B_c} - m_{J/\psi})^2 \approx (3180\,\mathrm{MeV})^2$). The $J/\psi\,\mu^+$ mass cut efficiencies for each set of generator level MC was compared to that of the nominal set, and the results are shown in Fig. 52. Thus the mass cut efficiency is quite sensitive to the pole mass used.

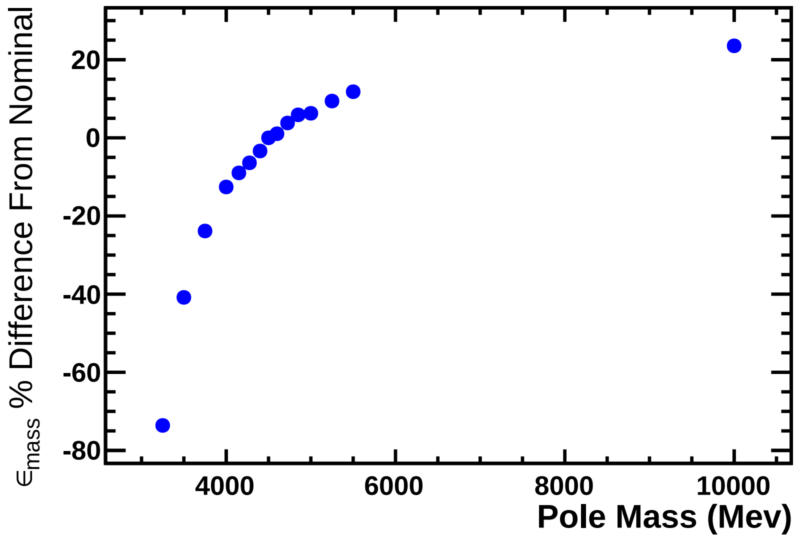


Figure 52: Percent difference from the nominal mass cut effiency ϵ_{mass} corresponding to $M(J/\psi\,\mu) > 5300\,\mathrm{MeV}$ for different values of the pole mass M_i

We compare the $M(J/\psi\,\mu)$ distributions of the fully simulated and nominal ($M_i = 4500\,\mathrm{MeV}$) generator level MC, as shown in Fig. 53. From this we see that the points are within error bars, and there is no significant $M(J/\psi\,\mu)$ efficiency dependence. Thus the fully simulated MC can be replaced with the generator level MC in the $M(J/\psi\,\mu)$ fit without any efficiency corrections to fix the shape.

The fits to the data were redone for each set of generator level MC. In these fits, we fix the signal shape before the fit to the data, rather than including it in a simultaneous fit since the statistical errors from the generator level sample sizes are small. For each fit the minimum χ^2 returned by the fit was plotted. The lowest value occurs for the nominal value with which the fully simulated MC was generated. The difference in χ^2 from the lowest value is shown for each set of generator level MC in Fig. 54.

From this we estimate that the pole mass which best fits the data is $M_i = 4500^{+850}_{-400}\,\mathrm{MeV}$. At both of these limits, the associated relative difference in ϵ_{mass} is $\sim 10\%$. This is larger than the observed variation between Kiselev's model and the other models. Therefore, we conclude that the observed endpoint shape of the $M(J/\psi\,\mu)$ distribution is not able to constrain the model parameters in a sensitive way. It is reassuring that the the value of the pole mass preferred by the data is the one chosen by the authors of the model.

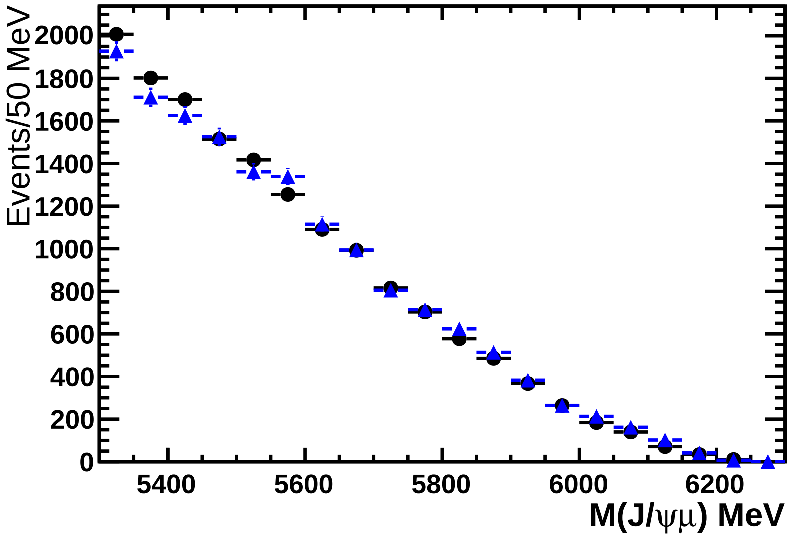


Figure 53: $M(J/\psi\,\mu)$ distributions of fully simulated MC (blue triangles) and generator level MC with the pole mass at the nominal value $M_i = 4500\,\mathrm{MeV}$ (black circles).

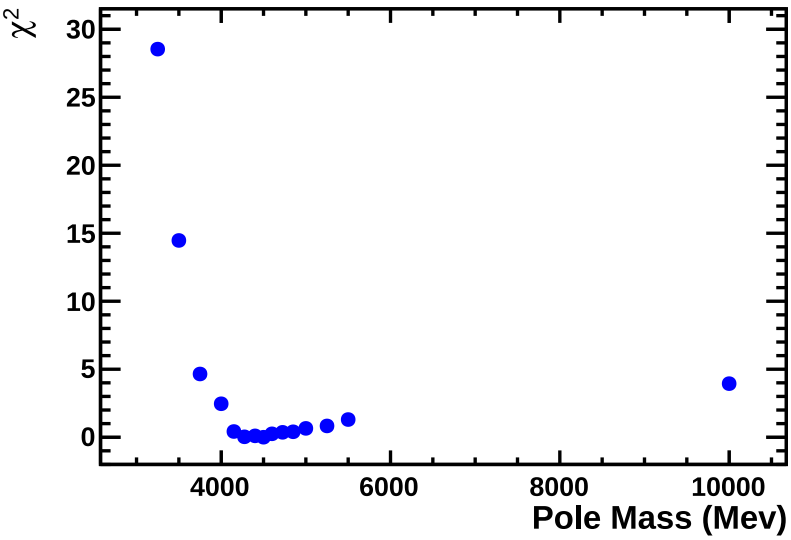


Figure 54: Difference in χ^2 from the minimum value for the fits to the $M(J/\psi\,\mu)$ distribution of the data with signal shapes coming from the MC generated with different M_i.

References

[1] CDF collaboration, F. Abe *et al.*, *Observation of the B_c Meson in $p\overline{p}$ Collisions at* $\sqrt{s} = 1.8$ TeV, Phys. Rev. Lett. **81** (1998) 2432, `arXiv:hep-ex/9805034`.

[2] CDF collaboration, A. Abulencia *et al.*, *Measurement of the B_c^+ Meson Lifetime Using the Decay Mode $B_c^+ \to J/\psi\,e^+\nu$*, Phys. Rev. Lett. **97** (2006) 012002,

`arXiv:hep-ex/0603027`.

[3] D0 collaboration, V. Abazov *et al.*, *Measurement of the Lifetime of the $B_c^\pm$ Meson in the Semileptonic Decay Channel*, Phys. Rev. Lett. **102** (2009) 092001, `arXiv:0805.2614`.

[4] CDF collaboration, A. Abulencia *et al.*, *Evidence for the exclusive decay $B_c^\pm \to J/\psi\,\pi^\pm$ and measurement of the mass of the $B_c^\pm$ meson*, Phys. Rev. Lett. **96** (2006) 082002, `arXiv:hep-ex/0505076`.

[5] D0 collaboration, V. Abazov *et al.*, *Observation of the B_c Meson in the Exclusive Decay $B_c \to J/\psi\,\pi$*, Phys. Rev. Lett. **101** (2008) 012001, `arXiv:0802.4258`.

[6] T. Nigmanov, K. Gibson, M. Hartz, and P. Shepard, *Measurement of B_c^+ properties at CDF*, DPF Conf. Proc (2009) `0910.3013`.

[7] A. Rakitin and S. Koshkarev, *Hadronic B_c Decays as a test of B_c cross section*, Phys. Rev. D. **81** (2010) 014005, `arXiv:0911.3287`.

[8] Particle Data Group, J. Beringer *et al.*, *Review of particle physics*, Phys. Rev. **D86** (2012) 010001.

[9] LHCb collaboration, R. Aaij *et al.*, *Measurements of B_c^+ production and mass with the $B_c^+ \to J/\psi\pi^+$ decay*, Phys. Rev. Lett. **109** (2012) 232001, `arXiv:1209.5634`.

[10] J. Wimberly, *b_c semileptonic form factors in evtgen*, Slides.

[11] B. Gui, T. Skwarnicki, and J. C. Wang, *Analysis of $B_{c,u}$ decays to J/ψ and three charged pions or kaons.*, LHCb-ANA-2011-054.

[12] LHCb collaboration, R. Aaij *et al.*, *First observation of the decay $B_c^+ \to J/\psi\,\pi^+\pi^-\pi^+$*, Phys. Rev. Lett. **108** (2012) 251802, `arXiv:1204.0079`.

[13] V. Kiselev, *Exclusive Decays and Lifetime of B_c Meson in QCD sum rules*, `arXiv:0211021`.

[14] V. Kiselev, A. Likhoded, and A. Onishchenko, *Semileptonic B_c meson decays in sum rules of QCD and NRQCD*, Nucl. Phys. **B569** (2000) 473, `arXiv:hep-ph/9905359`.

[15] V. Kiselev and A. Tkabladze, *Semileptonic B(c) decays from QCD sum rules*, Phys. Rev. **D48** (1993) 5208.

[16] D. Ebert, R. Faustov, and V. Galkin, *Weak decays of the B_c meson to charmonium and D mesons in the relativistic quark model*, Phys. Rev. D. **68** (2003) 094020, `arXiv:hep-ph/0306306v2`.

[17] D. Scora and N. Isgur, *Semileptonic Meson Decays in the Quark Model: An Update*, Phys. Rev. D. **52** (1995) 2783, `arXiv:hep-ph/9503486v1`.

[18] LHCb collaboration, R. Aaij *et al.*, *Measurement of the B_c^+ meson lifetime using $B_c^+ \to J/\psi\mu^+\nu_\mu X$ decays*, `1401.6932`.

[19] LHCb collaboration, R. Aaij *et al.*, *First observation of the decay $B_c^+ \to J/\psi K^+$*, `1306.6723`.

[20] C. Chang and Y. Chen, *Decays of the B_c meson*, Phys. Rev. D. **49** (1994) 3399.

[21] P. Colangelo and F. De Fazio, *Using heavy quark spin symmetry in semileptonic B_c decays*, Phys. Rev. D. **61** (1999) 034012, `arXiv:hep-ph/9909423v1`.

[22] J. Liu and K. Chao, *B_c meson weak decays and CP violation*, Phys. Rev. D. **56** (1997) 4133.

[23] H. Ke, T. Liu, and X. Li, *Transitions of $B_c \to \psi(1S, 2S)$ and the modified harmonic oscillator wave function in LFQM*, `arXiv:1307.5925`.

[24] X. Wang, W. Wang, and C. Lü, *B_c to p-wave Charmonia Transitions in Covariant Light-Front Approach*, Phys. Rev. D. **79** (2009) 114018, `arXiv:0901.1934`.

[25] M. Ivanov, J. Körner, and P. Santorelli, *Exclusive semileptonic and nonleptonic decays of the B_c meson*, Phys. Rev. D. **73** (2006) 054024, `arXiv:hep-ph/0602050v1`.

[26] C. Chang, Y. Chen, G. Wang, and H. Zong, *Decays of the meson B_c to a $P-$wave charmonium state χ_c or h_c*, Phys. Rev. D. **65** (2001) 014017, `arXiv:hep-ph/0103036v2`; C. Chang and Y. Chen, *Decays of the B_c meson*, Phys. Rev. D. **49** (1994) 3399.

[27] E. Hernandez, J. Nieves, and J. Verde-Velasco, *Study of exclusive semileptonic and nonleptonic decays of B_c^- in a nonrelativistic quark model*, Phys. Rev. D. **74** (2006) 074008, `arXiv:hep-ph/0607150v3`.

[28] A. Anisimov, I. Narodetskii, C. Semay, and B. Silvestre-Blac, *The B_c meson lifetime in the light-front constituent quark model*, Phys. Lett. B. **452** (1999) 129, `arXiv:hep-ph/9812514v1`; A. Anisimov, P. Kulikov, I. Narodetskii, and K. Ter-Martirosyan, *Exclusive and Inclusive Decays of the B_c Meson in the Light-Front ISGW Model*, Phys. Atom. Nucl. **62** (1999) 1739, `arXiv:hep-ph/9809249v1`.

[29] W. Wang, Y. Shen, and C. Lü, *Covariant light-front approach for B_c transition form factors*, Phys. Rev. D. **79** (2009) 054012, `arXiv:0811.3748`.

[30] W. Wang, Y. Fan, and Z. Xiao, *Semileptonic decays $B_c \to (\eta_c, J/\psi)l\nu$ in the perturbative QCD approach*, `arXiv:1212.5903`.

[31] F. Archilli *et al.*, *Performance of the Muon Identification at LHCb*, `arXiv:1306.0249`, LHCb-DP-2013-001.

[32] A. El-Hady, J. Muñoz, and J. Vary, *Semileptonic and non-leptonic B_c decays*, Phys. Rev. D. **62** (1999) 014019, `arXiv:hep-ph/9909406`.